AI BASED EMERGENCY RESOURCES ALLOCATION SYSTEM

A PROJECT REPORT

Submitted by

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in

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

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ABSTRACT

Emergency situations such as road accidents, fires, and medical emergencies require quick and efficient allocation of available emergency resources. Delays in selecting and dispatching suitable resources can increase response time and affect the outcome of an emergency. The proposed AI-Based Emergency Resource Allocation System is designed to support emergency response by automatically selecting suitable resources based on emergency type, priority, location, and resource availability.The system uses a knowledge-based approach in which predefined rules are used to determine the appropriate emergency resources such as ambulances, fire trucks, and police vehicles. The availability and suitability of resources are considered before allocation. To improve route selection, the system uses the A* search algorithm to identify an efficient path between the selected emergency resource and the incident location. Road network information and road conditions are considered during route planning.The system is implemented as a web-based application using a Python Flask backend with HTML, CSS, and JavaScript for the user interface. OpenStreetMap-based road network information is used for location and route-related processing. Security considerations such as user authentication, role-based access, input validation, and activity logging are included to reduce unauthorized access and improper modification of emergency information.The proposed system combines Intelligent Agents, Knowledge-Based Reasoning, Resource Allocation, and Heuristic Search into a single emergency response framework. The system aims to provide fast, explainable, and consistent resource allocation while reducing dependency on manual decision-making.

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1.INTRODUCTION

1.1BACKGROUND

Emergency situations require quick and effective allocation of suitable resources such as ambulances, fire trucks, and police vehicles. The proposed system uses knowledge-based reasoning and the A* search algorithm to allocate available emergency resources and identify an efficient route to the incident location.

1.2 PROBLEM STATEMENT

During emergency situations, selecting an appropriate response resource and reaching the incident location within a short time are critical. Traditional emergency response systems may depend on manual decision-making, which can result in delays in identifying suitable resources and planning efficient routes.

The AI-Based Emergency Resource Allocation System addresses this problem by providing an automated decision-support mechanism for emergency resource selection and route planning. The system uses predefined knowledge-based rules to determine the appropriate resource according to the emergency type and checks resource availability before allocation. After selecting a suitable resource, the A* search algorithm is used to determine an efficient route through the available road network.

The system also incorporates basic security mechanisms such as authentication, role-based access control, input validation, and allocation activity logging to protect the system from unauthorized access and maintain accountability.

1.3 OBJECTIVES

The main objective of the proposed AI-Based Emergency Resource Allocation System is to develop an intelligent decision-support system that can efficiently allocate emergency resources and identify suitable routes to emergency locations.

The specific objectives are:

1. To develop a knowledge-based reasoning system for emergency resource allocation using predefined IF-THEN rules.
2. To identify the appropriate emergency resource based on the type of emergency, such as an ambulance, fire truck, or police vehicle.
3. To check the availability of emergency resources before assigning a resource to an incident.
4. To use the A* search algorithm for finding an efficient route between the selected resource location and the emergency location.

5. To provide a map-based visualization of the emergency location, resource location, and calculated route.
6. To implement user authentication to restrict access to authorized users.
7. To implement Role-Based Access Control (RBAC) by providing different access permissions for Admin and Operator users.
8. To validate emergency and location inputs to reduce invalid or inappropriate requests.
9. To maintain allocation activity logs for monitoring and accountability.
10. To provide a simple and user-friendly interface for emergency resource allocation and route visualization.

1.4 SCOPE

The scope of the **AI-Based Emergency Resource Allocation System** is to provide an intelligent and secure decision-support mechanism for allocating emergency response resources based on predefined knowledge and available road-network information.

The system covers the following functionalities:

- Emergency registration based on predefined emergency types.
- Knowledge-based selection of suitable emergency resources.
- Resource availability checking.
- Location-based emergency resource allocation.
- A* based route calculation using a road network graph.
- Visualization of the calculated route using a digital map.
- Display of resource and route information to the user.
- User authentication using username and password.
- Role-based access for Admin and Operator users.
- Validation of emergency and geographical location inputs.
- Logging of resource allocation activities for administrative monitoring.

The system is intended as a decision-support prototype for demonstrating classical Artificial Intelligence techniques in emergency management. It does not replace real emergency control centres or human decision-makers. The effectiveness of the system depends on the availability and accuracy of resource and road-network data.

1.5 SECURITY REQUIREMENTS / SECURITY CONSIDERATIONS

Security is an important consideration because the system handles emergency information, resource allocation decisions, user accounts, and location information. The proposed system incorporates basic security controls to prevent unauthorized access and improve accountability.

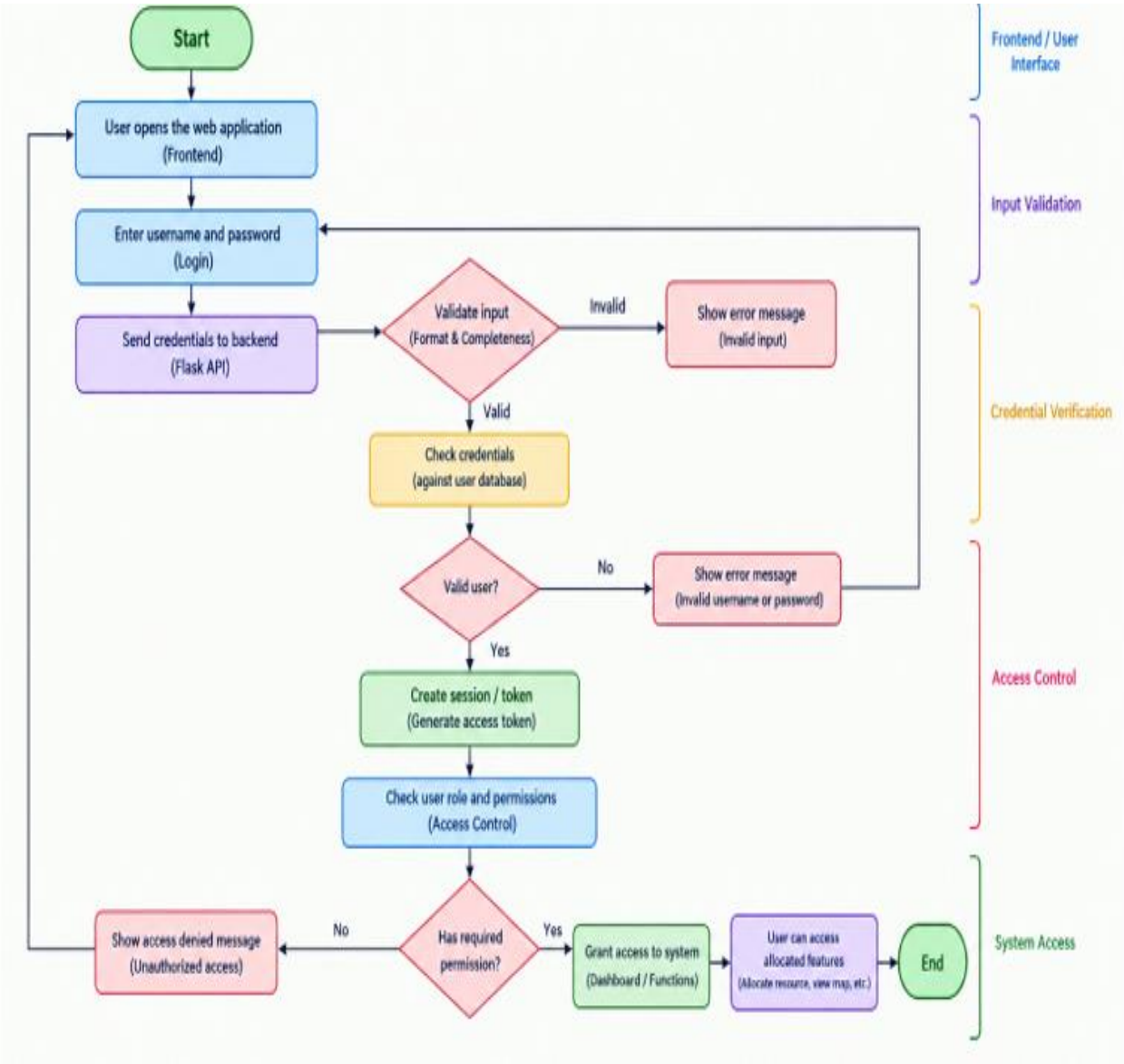


Figure 4.1: User Authentication and Access Control Flow

This flow represents a general security design. If the current project does not implement login, user accounts, or role-based permissions, this diagram should be treated as a proposed enhancement rather than an implemented module.

1.5.1 Authentication

The system requires users to authenticate using a valid username and password before accessing the emergency resource allocation dashboard. This prevents unauthorized users from directly accessing protected system functions.

1.5.2 Role-Based Access Control

The system provides two user roles:

- **Admin** – has access to administrative functions, including allocation activity logs.
- **Operator** – can perform emergency resource allocation and view the operational dashboard but cannot access administrative logs.

This ensures that users can access only the functions associated with their assigned role.

1.5.3 Input Validation

The system validates emergency input before processing an allocation request. Emergency types are checked against the supported values, while latitude and longitude values are validated to ensure that they contain valid numerical values within acceptable geographical ranges.

1.5.4 Password Protection

User passwords are not stored directly as plain text. Password hashing is used for storing user credentials, reducing the risk associated with direct exposure of passwords.

1.5.5 Access Control

Protected application routes require an authenticated session. Administrative routes additionally verify whether the logged-in user has the required Admin role.

1.5.6 Activity Logging

Successful resource allocation activities are recorded in the system. The log contains information such as emergency ID, user, emergency type, priority, allocated resource, location, timestamp, and allocation status. This provides accountability and helps administrators monitor system activities.

1.5.7 Secure Data Handling

The system follows basic secure data-handling practices by validating incoming information and limiting administrative information to authorized users. Location and emergency information are processed only for the intended allocation and routing functions.

1.6 SDG ALIGNMENT

1.6.1. SDG 3 – Good Health and Well-Being

- Relevant Target: Ensure healthy lives and promote well-being for all at all ages, including reducing preventable deaths and improving access to emergency healthcare services.

- Project Contribution: The proposed AI Emergency Resource Allocation System helps improve emergency response by identifying suitable emergency resources and calculating a route to the emergency location using the A* search algorithm.

1.6.2. SDG 9 – Industry, Innovation and Infrastructure

- Relevant Target: Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.
- Project Contribution: The project integrates Artificial Intelligence-based resource allocation, digital mapping, and graph-based pathfinding to support emergency service management.

1.6.3. SDG 11 – Sustainable Cities and Communities

- Relevant Target: Make cities and human settlements inclusive, safe, resilient, and sustainable.
- Project Contribution: The system supports urban emergency management by allowing users to select or search for an emergency location on a digital map. It identifies an appropriate available resource, when supported by the resource dataset, and displays the calculated route to the incident location. This can assist emergency service coordination in urban areas.

1.6.4. SDG Alignment Summary

SDG	**Goal**	**Project Contribution**
DG 3	Good Health and Well-Being	Supports faster emergency response through resource allocation and route calculation
SDG	Industry, Innovation and Infrastructure	Integrates AI-based decision support, digital mapping, and A* pathfinding.
SDG 11	Sustainable Cities and Communities	Supports urban emergency coordination and ambulance route planning.

Table 1.6.1. SDG Alignment

2.LITERATURE REVIEW

2.1 EXISTING SYSTEMS WITH SECURITY APPROACHES

Emergency response systems focus on selecting suitable emergency resources and directing them to incident locations within a short period. Existing research has explored shortest-path algorithms, ambulance routing, emergency resource allocation, and decision-support systems to improve emergency response efficiency.

Nordin et al. presented an ambulance routing approach using the A* algorithm for finding a shortest path between locations. Their work demonstrates the suitability of A* search for emergency vehicle routing because it can efficiently determine a path between a source and destination in a road network. This provides the basis for using A* in the proposed system for route planning.

Barrachina et al. studied emergency response and communication approaches for improving the arrival time of emergency vehicles. Their work highlights the importance of efficient routing and communication in emergency situations, where reducing response time is an important requirement.

Pettet et al. investigated decision-support approaches for emergency response operations. Their work focuses on supporting emergency response decisions involving resources and incidents. Such approaches demonstrate the importance of automated decision support when multiple emergency resources need to be considered.

Existing emergency response research also considers resource allocation and dispatch as important components of emergency management. These approaches show that combining resource selection with route planning can support faster and more systematic emergency response.

From a security perspective, an emergency resource allocation system must also prevent unauthorized users from accessing or modifying operational information. Authentication, role-based access control, input validation, and activity logging can be incorporated to provide basic protection and accountability.

2.2 RESEARCH GAPS

Although existing studies have addressed emergency routing, resource allocation, and emergency decision support, several gaps remain when these concepts are considered together in a simple classical AI-based system.

The major research gaps identified are:

- **Separate consideration of resource allocation and routing:**
  Some approaches primarily focus on routing, while others focus on resource allocation. A combined system that first selects an appropriate available resource and then calculates an efficient route can provide more complete decision support.
- **Limited use of simple knowledge-based reasoning:**
  Many modern emergency systems can involve complex optimization or data-driven approaches. There is a need for simpler classical AI approaches that can be implemented and understood easily using predefined IF-THEN rules.
- **Resource availability consideration:**
  Selecting a resource based only on emergency type is insufficient. The availability of the selected emergency resource should also be checked before allocation.
- **Security and accountability:**
  Emergency systems handle operational information and therefore require controlled access. Basic authentication, role-based access control, input validation, and allocation logging should be integrated with the allocation process.
- **Map-based decision support:**
  Presenting the selected resource and calculated route on an interactive map can make the system easier for an emergency operator to understand and use.

The proposed system addresses these gaps by combining knowledge-based reasoning, resource availability checking, A* route planning, map visualization, and basic security controls in a single system.

2.3 PROPOSED SYSTEM

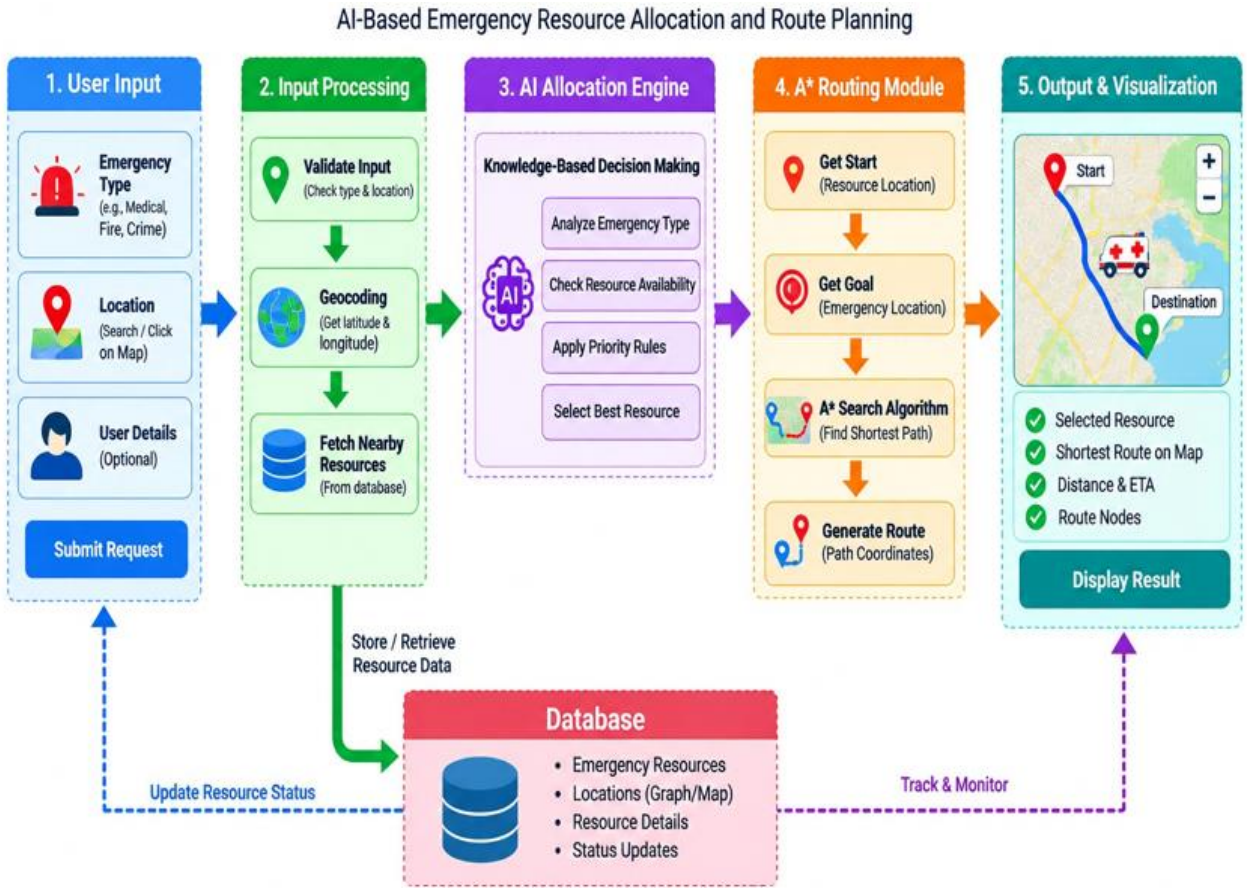


Figure 2.3: Proposed system

The proposed **AI-Based Emergency Resource Allocation System** is a classical AI-based decision-support system designed to assist in selecting suitable emergency resources and determining an efficient route to an emergency location.

The system uses predefined knowledge-based rules to associate emergency types with appropriate resources. For example, an accident or medical emergency can require an ambulance, a fire emergency can require a fire truck, and a crime-related emergency can require a police vehicle.

After identifying the required resource type, the system checks the availability of the resource. Once a suitable resource is identified, the A* search algorithm is applied to the road network to determine an efficient route between the resource location and the emergency location.

The system provides the allocation result and route information through a web-based dashboard. Security mechanisms such as user authentication, role-based access control, input validation, password hashing, and allocation logging are also incorporated.

2.3.1 OVERVIEW OF PROPOSED SYSTEM

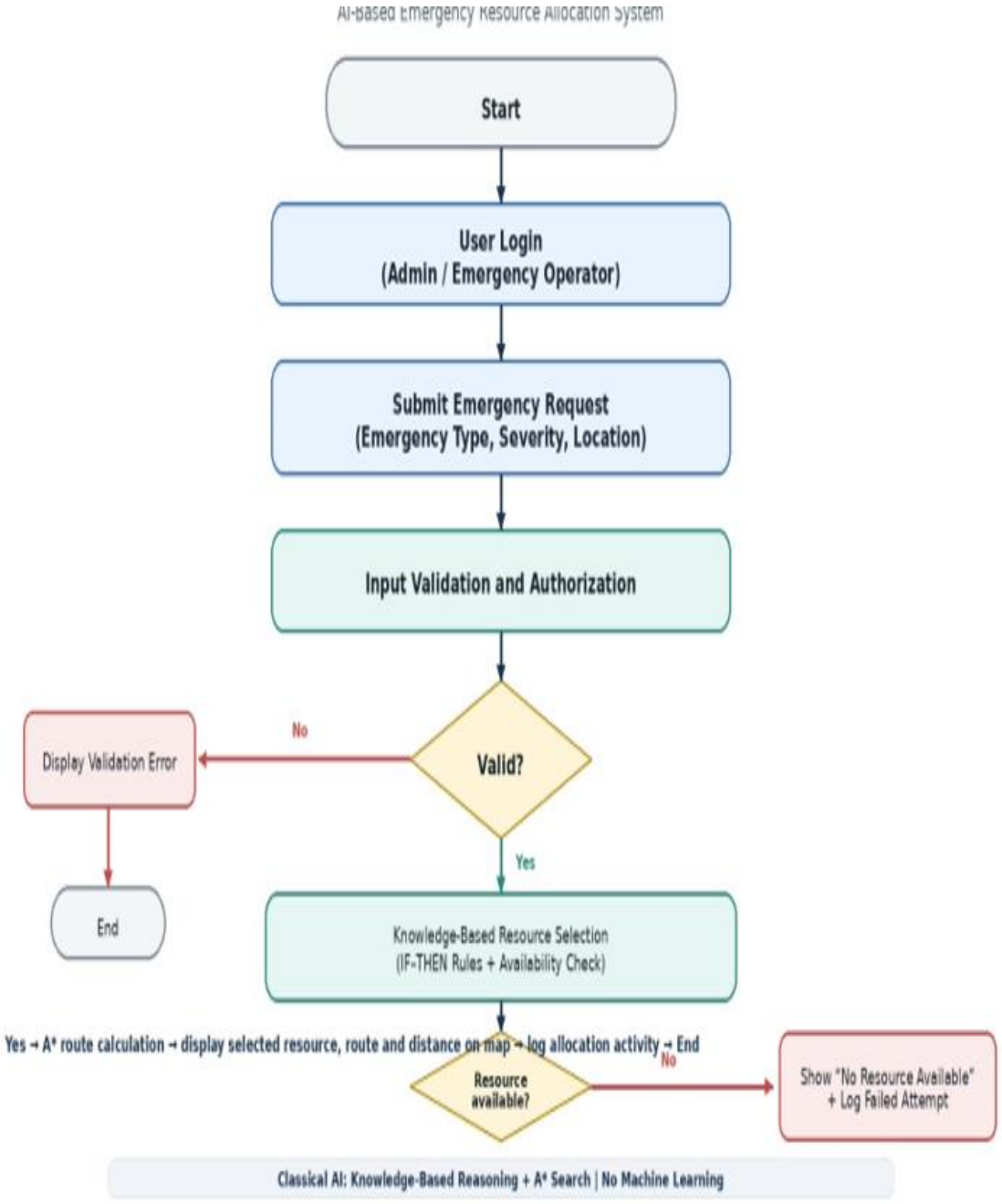


Figure 2.3.1: Workflow of Proposed Diagram

The proposed system consists of several interconnected components.

First, the user logs into the system using valid credentials. Based on the authenticated user's role, access to system functions is provided. The system supports **Admin** and **Operator** roles.

The operator can enter the emergency type and emergency location. The system validates the received information before processing it. The knowledge-based reasoning module then determines the appropriate emergency resource.

The resource allocation module checks whether a suitable resource is available. If an available resource is found, its location is obtained and passed to the routing module.

The routing module uses the *A search algorithm** with the road network to determine an efficient route from the selected resource to the emergency location. The calculated route is displayed on an interactive map.

The allocation activity is also recorded in the system log. This provides traceability of emergency allocation operations and allows authorized administrators to review previous allocation activities.

The overall process can be represented as:

*User Login - Input Validation - Knowledge-Based Reasoning -Resource Availability Check - Resource Allocation - A Route Calculation - Map Display - Activity Logging**

2.3.2 OBJECTIVES OF PROPOSED SYSTEM

The proposed system is developed with the following objectives:

1. To automate the selection of emergency resources using knowledge-based reasoning.
2. To associate different emergency types with suitable emergency resources.
3. To verify the availability of emergency resources before allocation.
4. To use the A* search algorithm for efficient emergency route planning.
5. To display the selected emergency resource and calculated route on an interactive map.
6. To provide a simple web-based interface for emergency operators.
7. To provide authentication for authorized system users.
8. To implement role-based access control for Admin and Operator users.
9. To validate emergency and geographical input data.
10. To maintain allocation activity logs for accountability and administrative monitoring.

2.3.3 ADVANTAGES OF PROPOSED SYSTEM

The proposed system provides the following advantages:

1. Automated Resource Selection

The knowledge-based reasoning module automatically selects an appropriate resource according to the emergency type, reducing the need for manual resource selection.

2. Availability-Based Allocation

The system checks resource availability before assigning a resource. This helps avoid allocating resources that are already unavailable.

3. Efficient Route Planning

The A* algorithm is used to calculate an efficient path between the emergency resource and the incident location.

4. Interactive Map Visualization

The selected resource and calculated route are displayed on a map, making the result easier for the operator to understand.

5. Faster Decision Support

By combining resource selection and route planning, the system can provide the operator with the required information through a single dashboard.

6. Basic Security

Authentication and role-based access control restrict system functions to authorized users.

7. Input Validation

Emergency and location information is validated before it is processed by the allocation system.

8. Activity Logging

Allocation operations are recorded with details such as emergency ID, user, emergency type, resource, location, timestamp, and allocation status.

9. Simple Classical AI Approach

The system uses understandable *IF-THEN knowledge-based rules and A search**, making the AI methodology suitable for an academic classical AI project without requiring machine learning or neural networks.

10. Decision-Support System

The proposed system assists emergency operators in making resource allocation and routing decisions. It is intended as a decision-support prototype and does not replace human emergency response authorities.

3. SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

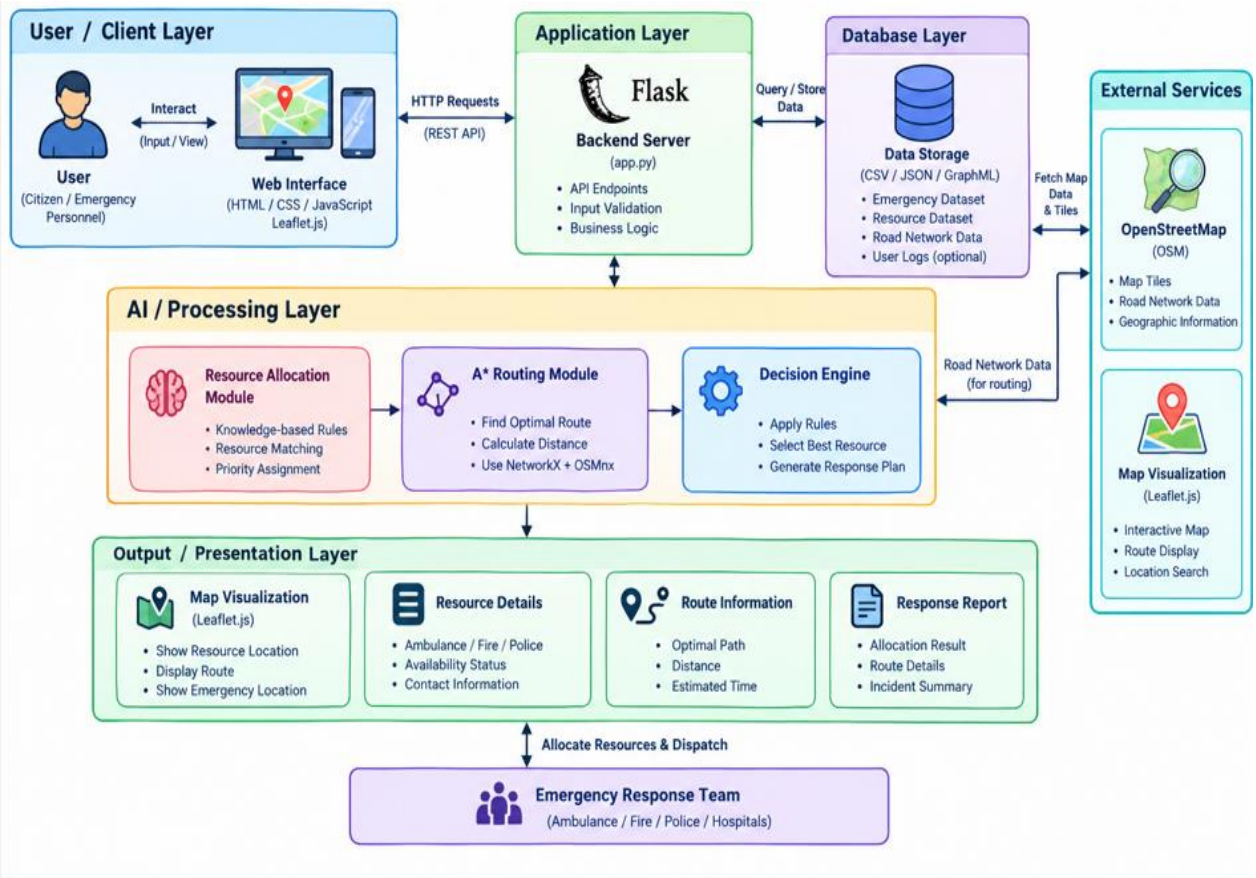


Figure 3.1:System Architecture

The **AI-Based Emergency Resource Allocation System** is designed as a web-based decision-support system that combines classical AI techniques with basic security mechanisms. The system accepts emergency information from an authorized user, identifies a suitable emergency resource using knowledge-based reasoning, checks resource availability, and calculates an efficient route using the A* search algorithm.

The system architecture consists of the following major layers:

1. **User Interface Layer**
2. **Authentication and Security Layer**
3. **Knowledge-Based Reasoning Layer**
4. **Resource Allocation Layer**
5. *A Route Planning Layer**
6. **Database and Logging Layer**
7. **Map Visualization Layer**

3.1.1 User Interface Layer

The user interface provides a web-based dashboard through which an authorized operator can enter emergency details and view the allocation result.

The interface provides:

- Emergency type selection
- Emergency location selection
- Current location detection
- Interactive map
- Selected resource information
- Resource location
- Route distance
- Route node count
- Calculated route visualization
- User and role information

The interface is implemented using HTML, CSS, JavaScript, and Leaflet for map visualization.

3.1.2 Authentication and Security Layer

The authentication layer ensures that only authorized users can access the protected system functions.

The system supports two roles:

Role	Main Permissions
Admin	View allocation logs and administrative information
Operator	Report emergencies and request resource allocation

User credentials are stored using password hashing rather than plain-text passwords. Session-based authentication is used to maintain the user's login state.

3.1.3 Knowledge-Based Reasoning Layer

The knowledge-based reasoning module uses predefined rules to determine the suitable resource according to the emergency type.

Emergency Type	Required Resource
Accident	Ambulance
Medical	Ambulance
Fire	Fire Truck
Crime	Police Vehicle

3.1.4 Resource Allocation Layer

After identifying the required resource type, the system checks the available resources.

The allocation process considers:

- Emergency type
- Required resource type
- Resource availability
- Resource location

If a suitable resource is available, it is selected for the emergency.

3.1.5 A* Route Planning Layer

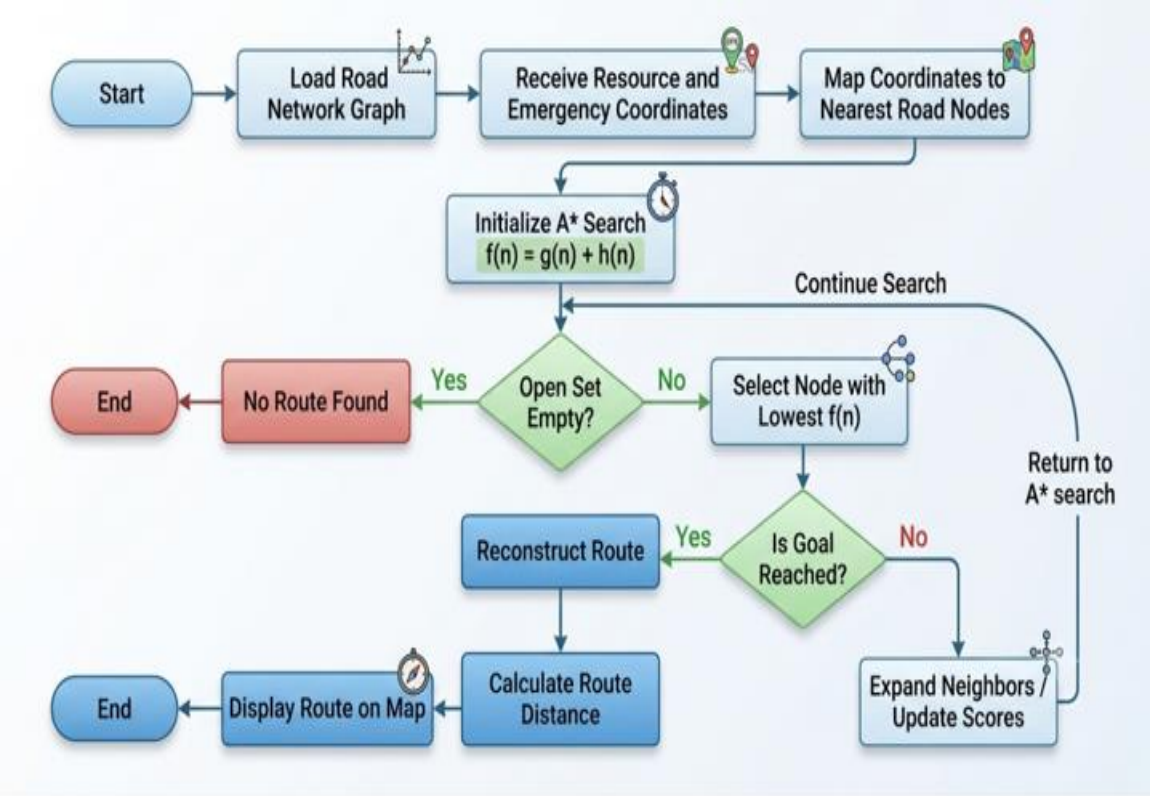


Figure 3.1.5 : A* Route Planning Flowchart

The selected resource location and emergency location are provided to the A* routing module. The A* algorithm searches the road network and determines an efficient path between the two locations.

The algorithm uses:

- Current node
- Goal node
- Path cost
- Heuristic cost
- Road distance

The total evaluation function is:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$= actual cost from the starting node to node n
- $h(n)$= estimated cost from node $n$to the destination
- $f(n)$= total estimated cost

The resulting route is then displayed on the map.

3.1.6 Database and Logging Layer

SQLite is used for storing system security and allocation information.

The security database stores:

- Username
- Password hash
- User role
- Emergency allocation logs

Each allocation activity can contain:

- Emergency ID
- Username
- Emergency type
- Priority
- Resource ID
- Emergency location
- Timestamp
- Allocation status

This provides basic accountability for resource allocation activities.

3.1.7 Map Visualization Layer

The calculated A* route is displayed using an interactive map. The map shows the emergency location, selected resource location, and route connecting the two locations.

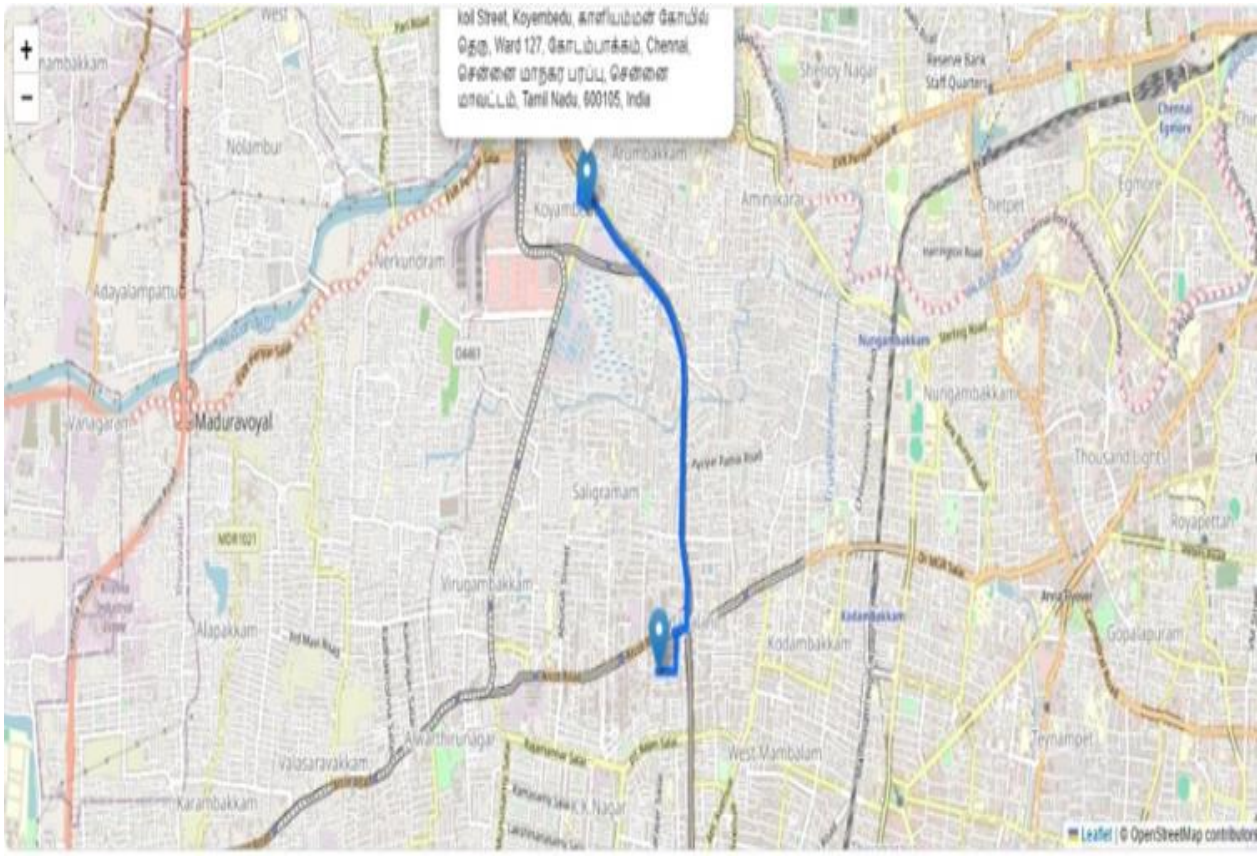


Figure 3.1.7: Map Visualization Layer

The map component helps the operator visually understand the result of the allocation and route-planning process.

3.2 MODULE DESIGN

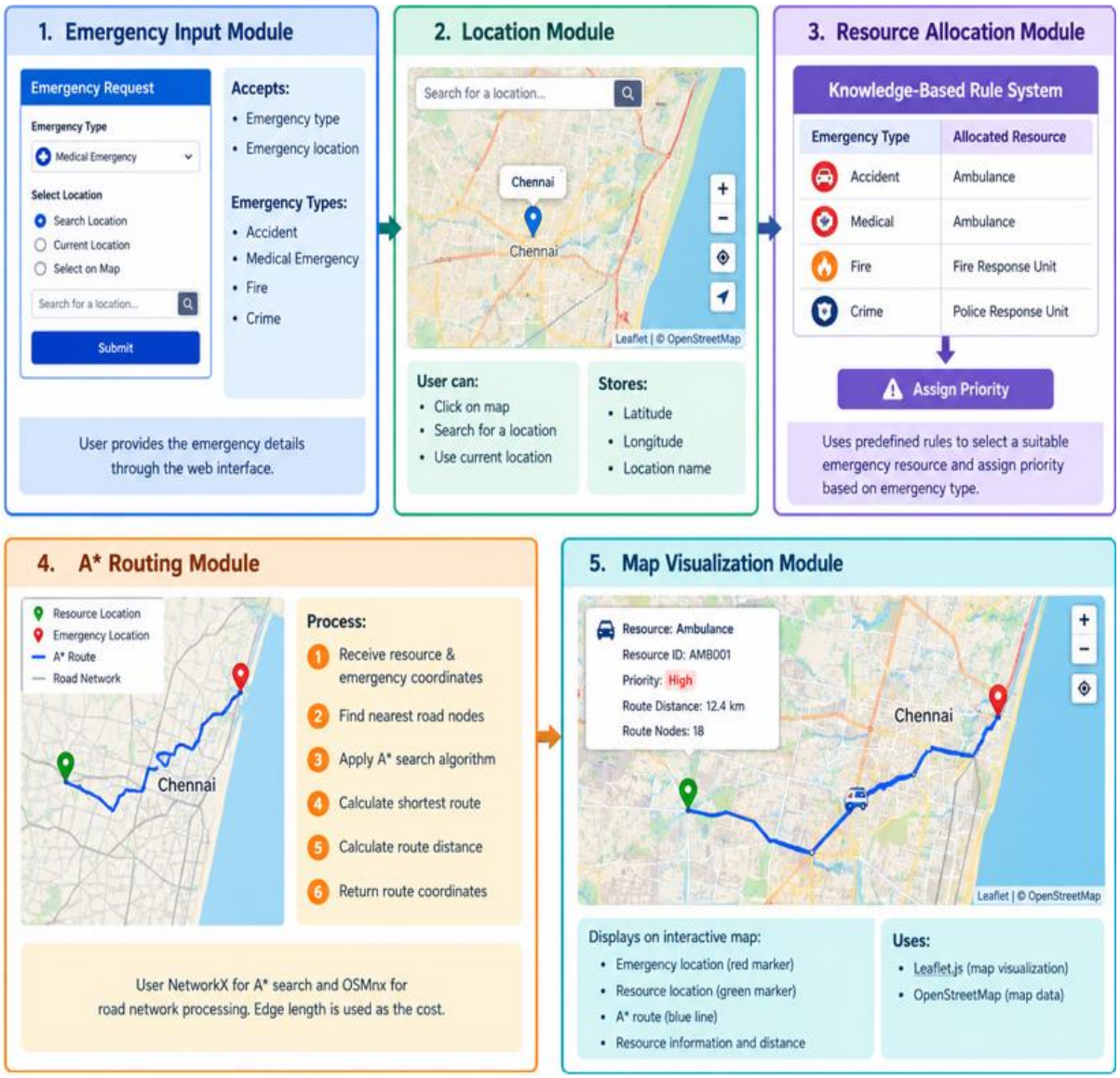


Figure 3.2:Module Design

The proposed system is divided into multiple modules to make the implementation organized and maintainable.

3.2.1 Authentication Module

The authentication module verifies the username and password provided by the user.

Input	Username and password
Processing	Credential verification
Output	Successful login or authentication failure

The module also establishes a user session after successful authentication.

3.2.2 Role-Based Access Control Module

The RBAC module determines the functions available to the authenticated user.

The system provides two roles:

Admin

- Access administrative information
- View allocation logs
- Monitor allocation activities

Operator

- Enter emergency details
- Request resource allocation
- View allocation results
- View calculated routes

Unauthorized access to administrative functions is rejected by the system.

3.2.3 Input Validation Module

The input validation module verifies emergency information before it is processed.

The system validates:

- Emergency type
- Latitude
- Longitude
- Required input fields

Latitude values must be within:

$$-90 \leq Latitude \leq 90$$

Longitude values must be within:

$$-180 \leq Longitude \leq 180$$

Invalid values are rejected to prevent incorrect allocation requests.

3.2.4 Knowledge-Based Resource Allocation Module

This module contains the predefined emergency-resource rules.

The module receives the emergency type and determines the appropriate resource category.

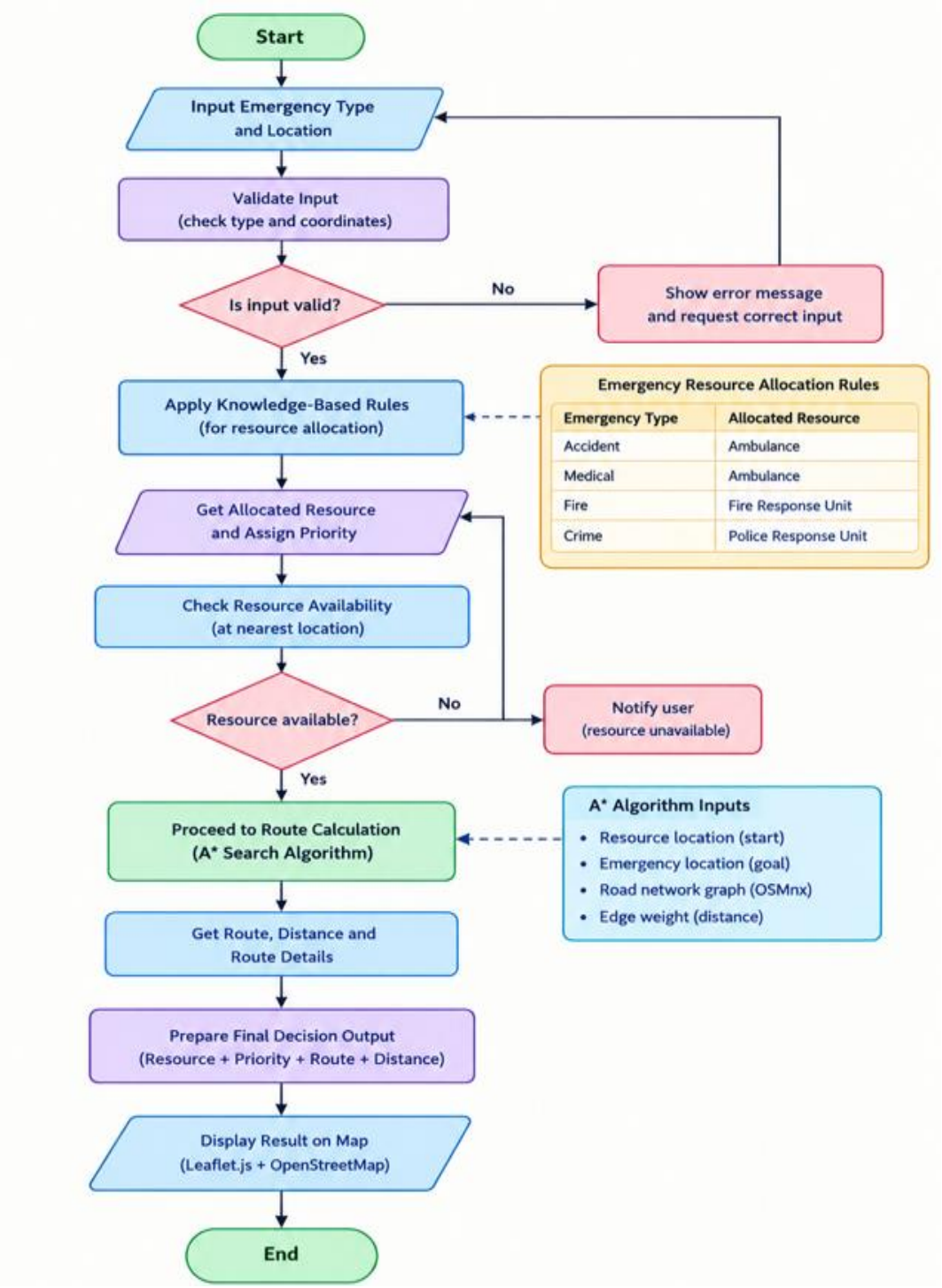


Figure 4.3: Knowledge-Based Emergency

Example

Emergency Type	Required Resource
Accident	Ambulance
Medical	Ambulance
Fire	Fire Truck
Crime	Police Vehicle

Table 3.4 :Example

3.2.5 Resource Availability Module

The resource availability module determines whether the required emergency resource can be assigned.

If the required resource is available, its information is returned to the main application.

The information may include:

- Resource ID
- Resource type
- Resource location
- Availability status

If no suitable resource is available, the system returns an appropriate allocation failure message.

3.2.6 A* Search Module

The A* module is responsible for route calculation.

The module receives:

- Road network
- Resource latitude
- Resource longitude
- Emergency latitude
- Emergency longitude

The nearest road-network nodes corresponding to the resource and emergency locations are identified. A* search is then applied to calculate the route.

The module returns:

- Route coordinates
- Route distance
- Number of nodes in the calculated route

3.2.7 Map Visualization Module

The map visualization module displays the emergency location and selected resource location. The calculated route is represented graphically on the map. This allows the user to understand the route selected by the A* algorithm.

3.2.8 Activity Logging Module

The activity logging module records successful allocation operations.

Each log contains an automatically generated emergency ID along with user and allocation information.This module provides accountability and allows administrators to review previous allocation activities.

3.3 DATA FLOW / UML DIAGRAMS

3.3.1 Use Case Description

The system has two primary actors:

- **Admin**
- **Operator**

The Operator interacts with the system to perform emergency resource allocation.

The Admin can access allocation logs and administrative information.

Main use cases include:

- Login
- Enter emergency details
- Validate emergency information
- Request resource allocation
- Check resource availability
- Calculate A* route
- View allocation result
- View route on map
- View allocation logs
- Logout

Level 0 DFD

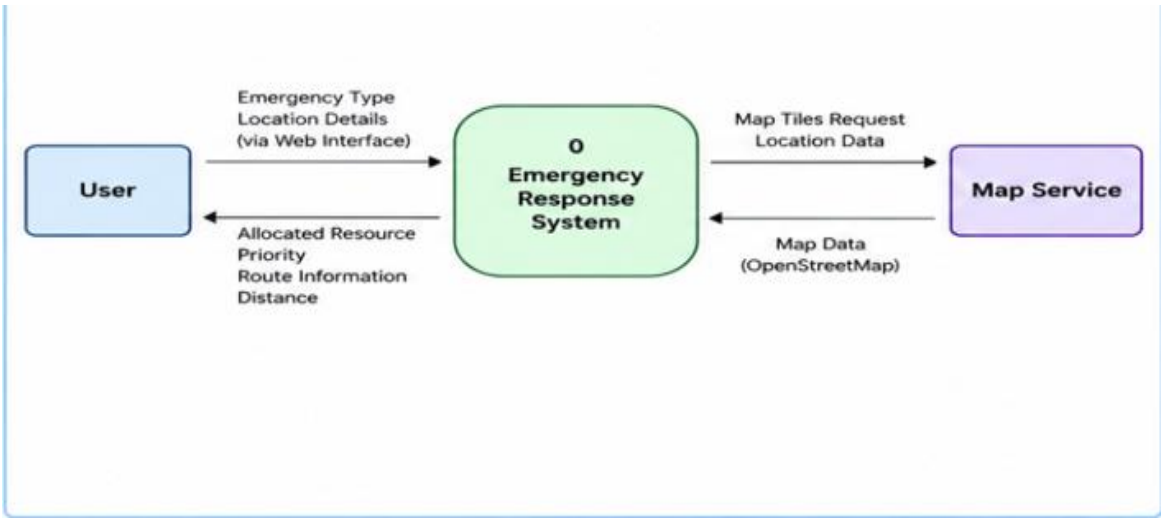


Figure 3.3.1 Level 0 DFD

Level 1 DFD

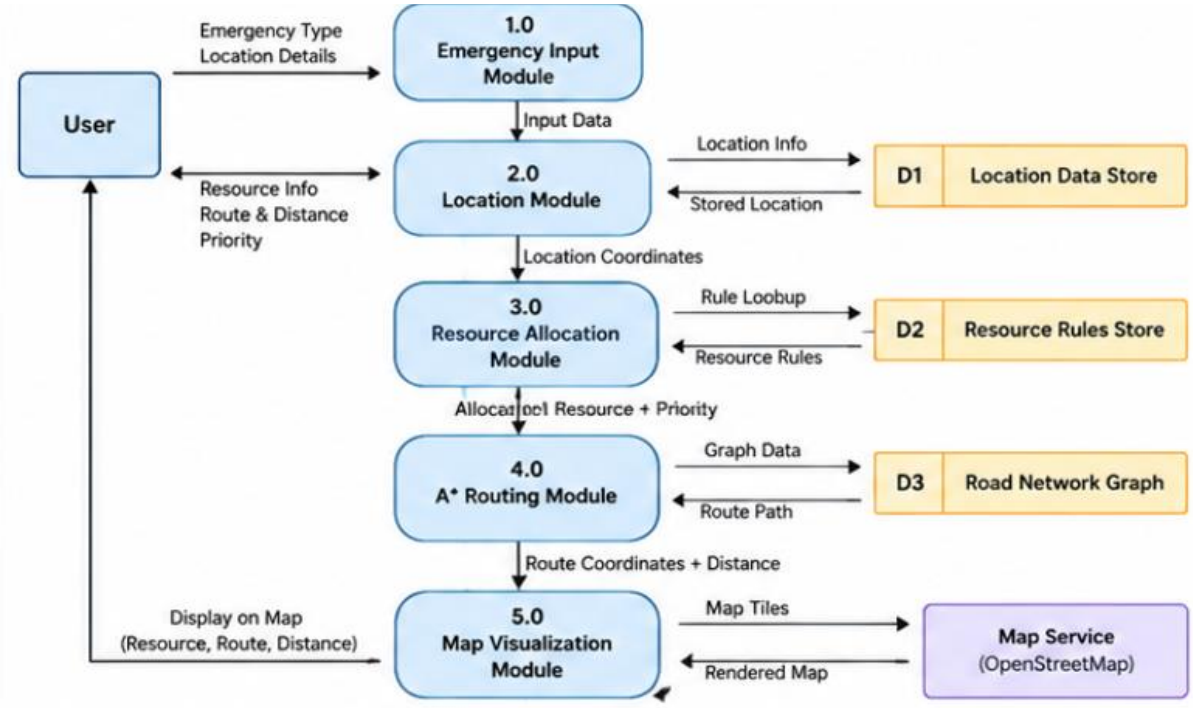


Figure 3.3.1:Level 1 DFD

3.3.2 USE CASE DIAGRAM

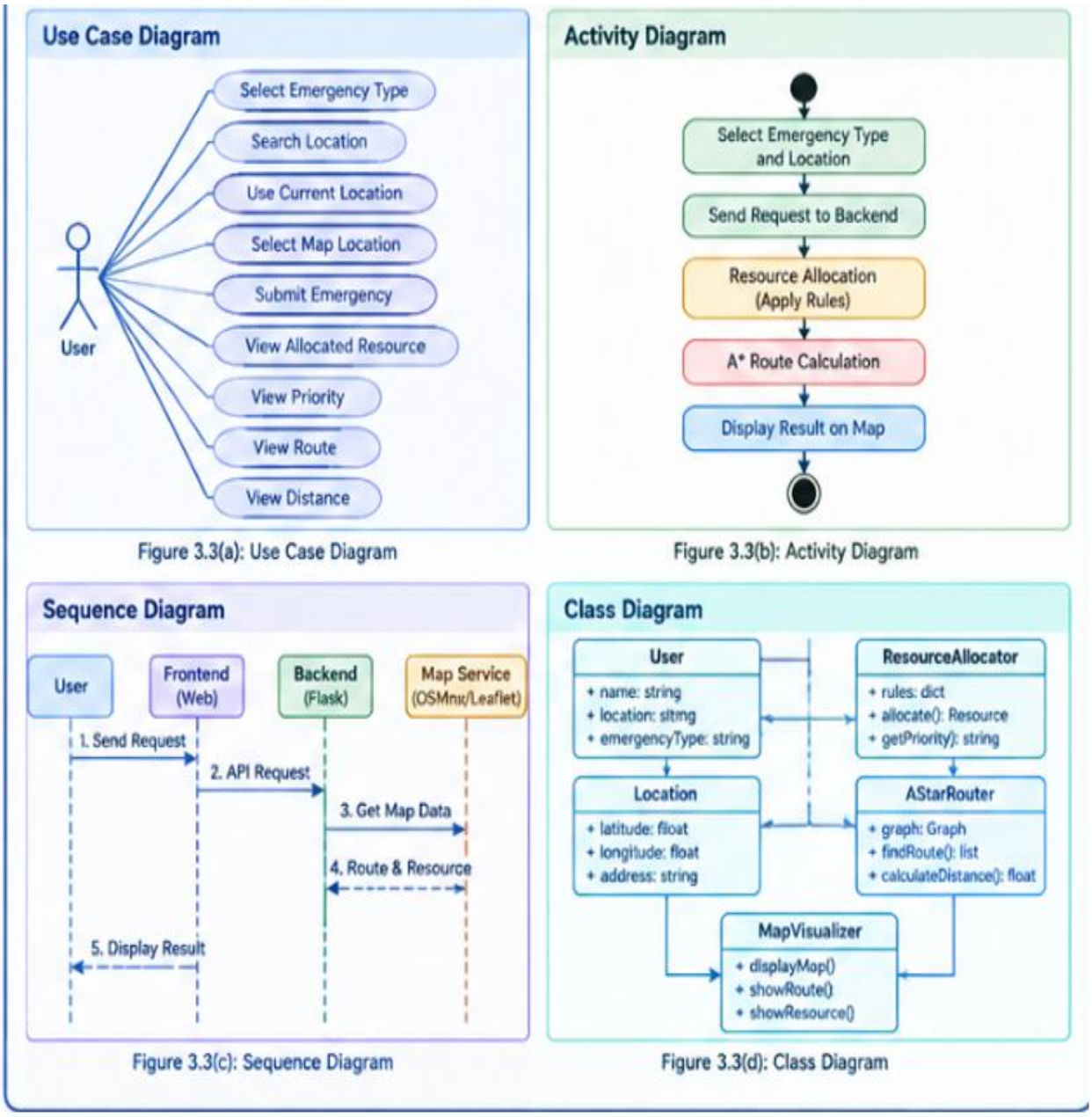


Figure 3.3(a): Use Case Diagram

Figure 3.3(b): Activity Diagram

Figure 3.3(c): Sequence Diagram

Figure 3.3(d): Class Diagram

Figure 3.3.2:Use diagram

3.3.3 SEQUENCE DIAGRAM FOR EMERGENCY HANDLING

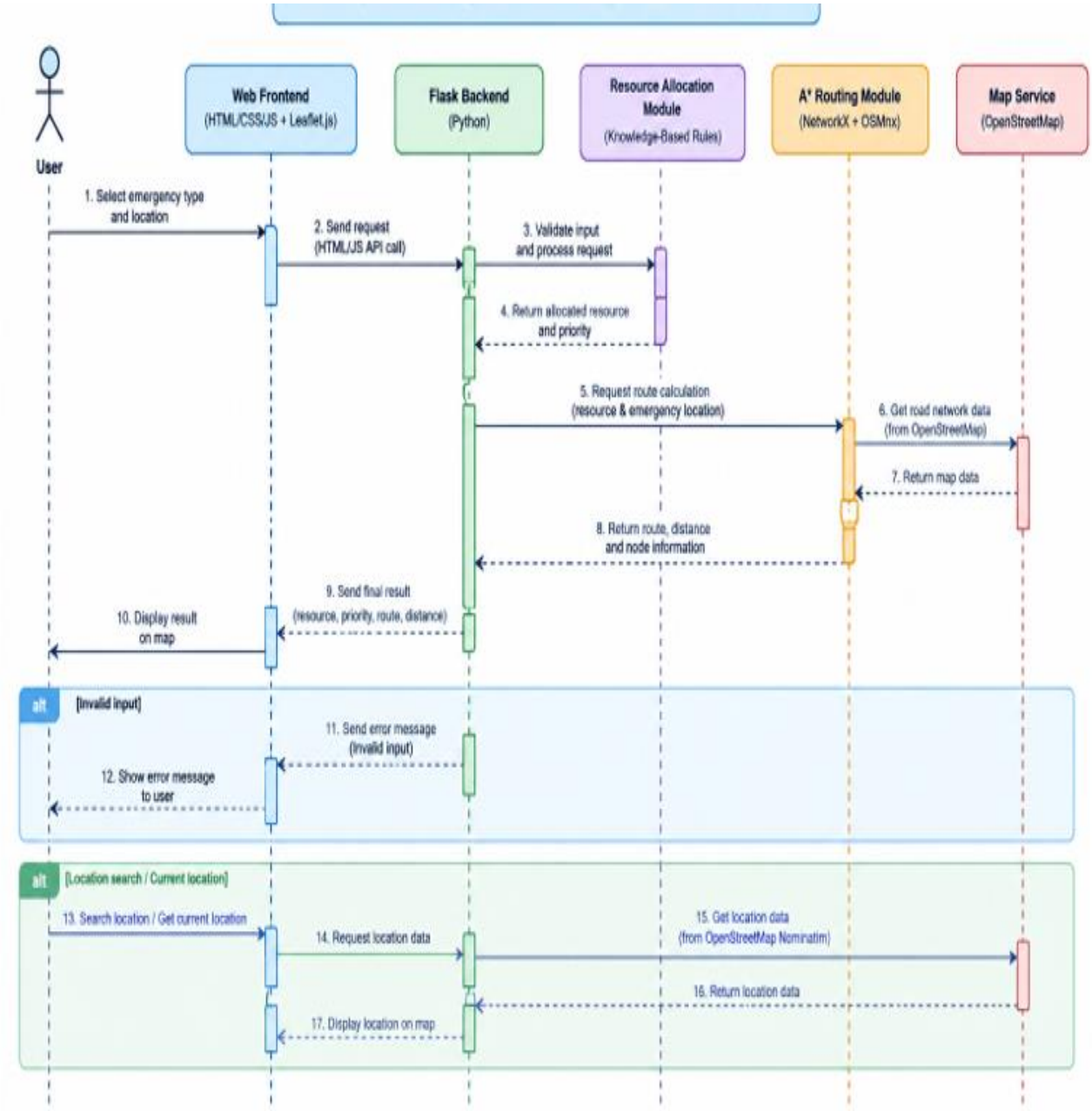


Figure 3.3.3: Sequence Diagram for Emergency Handling

The sequence diagram illustrates the interaction between the user, web frontend, Flask backend, resource allocation module, A* routing module, and map service during emergency handling. Initially, the user selects the emergency type and location through the web interface. The frontend sends the request to the Flask backend, which validates the input and invokes the resource allocation module.

3.3.4 ACTIVITY DIAGRAM FOR EMERGENCY RESOURCE ALLOCATION

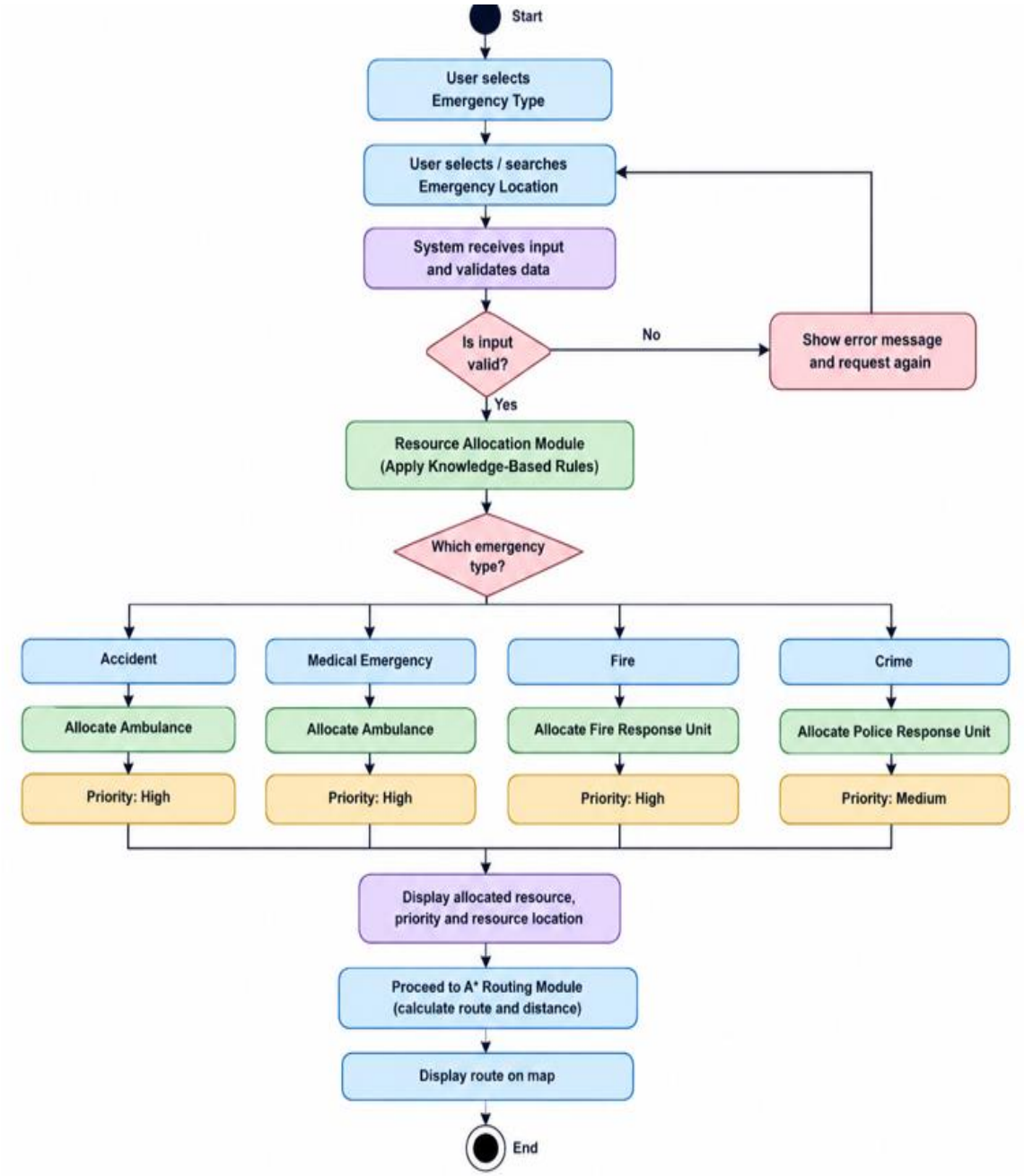


Figure 3.3.4: Activity Diagram for Emergency Resource Allocation

The activity diagram illustrates the workflow of the proposed AI Emergency Resource Allocation System. Initially, the user selects the emergency type and location. The system validates the entered information and displays an error message if the input is invalid.

3.4 THREAT MODEL AND ATTACK SURFACE

The proposed system contains several interfaces through which unauthorized access or invalid data could potentially affect the system.

3.4.1 Login Interface

The login interface accepts username and password information.

Potential threat: Unauthorized users attempting to access the system.

Security control: Authentication and password hashing.

3.4.2 Emergency Input Interface

The emergency form accepts emergency type and geographical coordinates.

Potential threat: Invalid or manipulated input.

Security control: Input validation and range checking.

3.4.3 Allocation API

The backend allocation API processes emergency requests and returns resource information.

Potential threat: Unauthorized requests or invalid data being submitted directly to the API.

Security control: Session authentication and server-side input validation.

3.4.4 Administrative Interface

The administrative functionality provides access to allocation logs.

Potential threat: An Operator attempting to access administrator-only information.

Security control: Role-based access control.

3.4.5 Allocation Logs

Allocation logs contain information about emergency requests and selected resources.

Potential threat: Unauthorized access to operational information.

Security control: Restrict log access to authorized Admin users.

3.5 SECURITY CONTROLS

The following security controls are implemented in the proposed system.

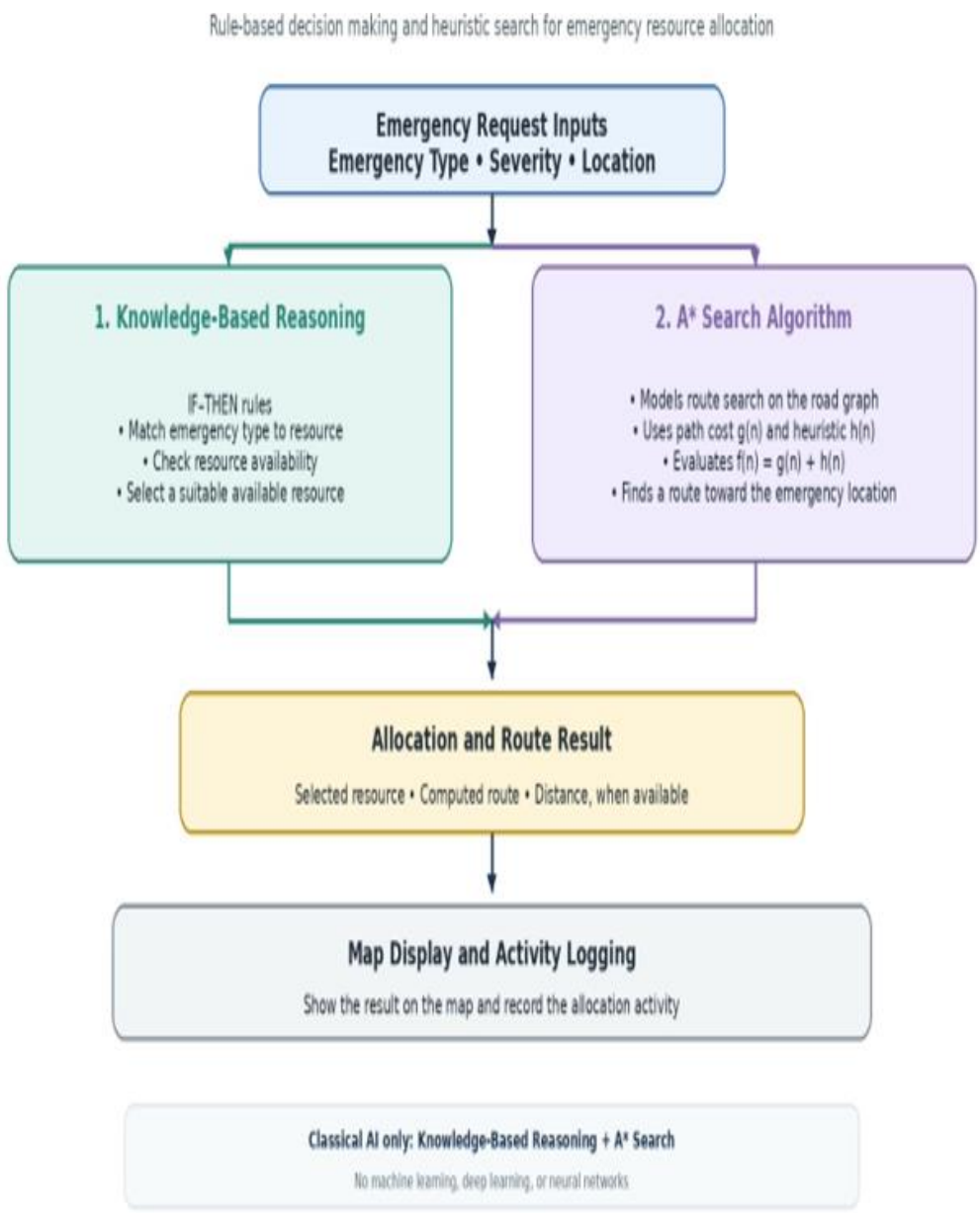


Figure 3.4: Classical AI Techniques Used in the System

3.5.1 Authentication

Users must provide valid credentials before accessing the main emergency resource allocation dashboard.

3.5.2 Password Hashing

Passwords are stored as password hashes rather than plain-text passwords. This provides basic protection for stored credentials.

3.5.3 Role-Based Access Control

The system separates users into Admin and Operator roles. Administrative functions are restricted to authorized Admin users.

3.5.4 Input Validation

The system validates emergency type and geographical coordinates before processing an allocation request.

3.5.5 Session-Based Access Control

Protected backend operations require an authenticated user session.

3.5.6 Allocation Activity Logging

Allocation operations are recorded with emergency and user information. This provides accountability and allows administrators to monitor previous activities.

3.5.7 Secure Data Handling

The system uses controlled database access and parameterized SQL queries for storing and retrieving security and allocation information.

3.6 HARDWARE AND SOFTWARE REQUIREMENTS

3.6.1 Hardware Requirements

Component	**Requirement**
Processor	Intel Core i3 or equivalent and above
RAM	Minimum 4 GB; 8 GB recommended
Storage	Minimum 10 GB free space
Display	Standard monitor
Internet	Required for online map and location services

3.6.2 Software Requirements

Software	**Purpose**
Windows 10/11	Operating System
Python 3.12.3	Application development
Flask	Web backend
HTML/CSS/JavaScript	Frontend development
Leaflet.js	Interactive map visualization
OSMnx	Road network processing
NetworkX	Graph and A* processing
scikit-learn	Required dependency for the Python routing environment
SQLite	Security and allocation data storage
Web Browser	System access and visualization
OpenStreetMap	Map and road-network data

4. SYSTEM DESCRIPTION

4.1 DATA COLLECTION

The proposed **AI-Based Emergency Resource Allocation System** requires emergency, resource, location, and road-network information for performing resource allocation and route planning.

The main data used by the system are:

- Emergency type
- Emergency location
- Resource type
- Resource ID
- Resource availability
- Resource location
- Road network information
- User and role information
- Allocation activity information

4.1.1 Emergency Data

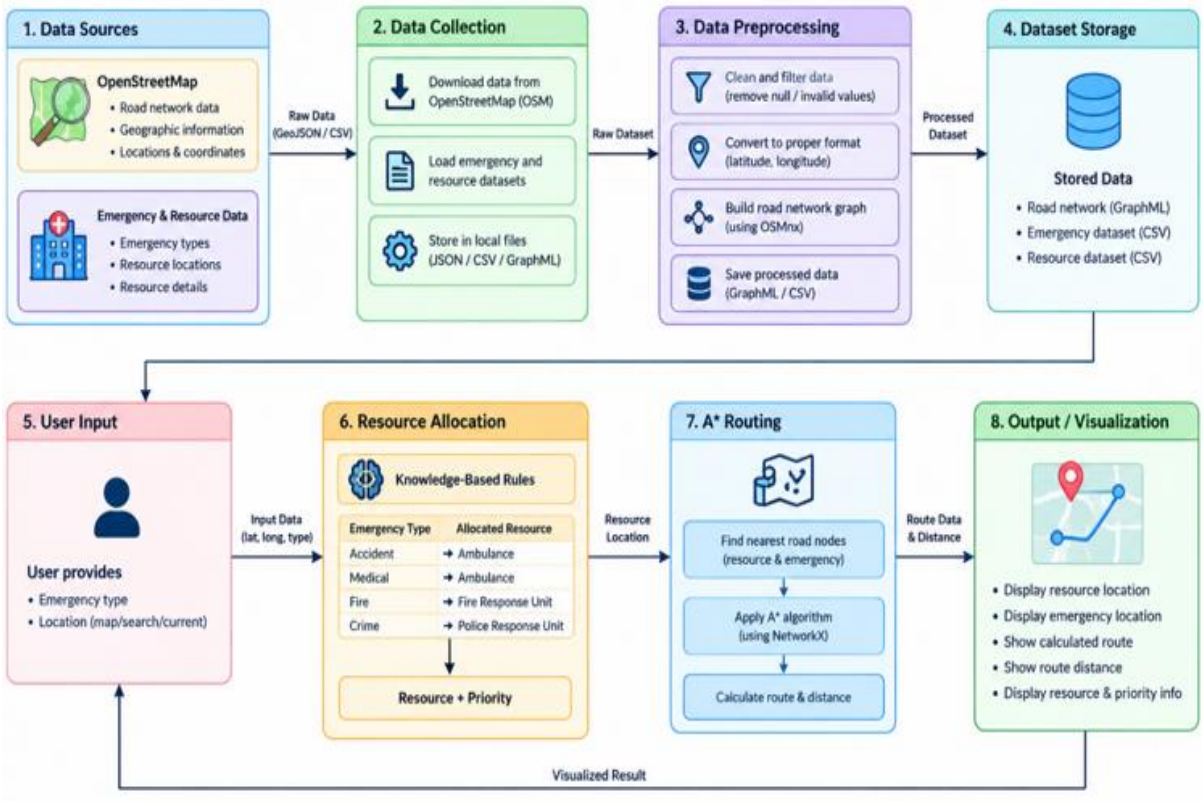


Figure 4.2: Emergency and Resource Dataset Flow Diagram

The system considers different emergency categories and maps them to suitable emergency resources.

Emergency Type	Required Resource
Accident	Ambulance
Medical Emergency	Ambulance
Fire	Fire Truck
Crime	Police Vehicle

The emergency location is represented using latitude and longitude coordinates.

4.1.2 Resource Data

The resource information contains details required for selecting an appropriate emergency vehicle.

Typical resource attributes include:

- Resource ID
- Resource type
- Location
- Availability status

The availability status is checked before the resource is allocated.

4.1.3 Road Network Data

The system uses a road network graph for route calculation. The road network contains nodes and edges representing locations and road connections.

The road network is used by the A* search algorithm to determine an efficient route from the selected emergency resource to the incident location.

4.1.4 Security Data

The security component maintains user information required for authentication and authorization.

The system supports:

- Admin
- Operator

User passwords are stored in hashed form, and authenticated sessions are used to protect the application.

4.2 DATA PREPROCESSING

Before emergency information is processed, the system performs basic validation and preparation.

4.2.1 Emergency Type Validation

The emergency type entered by the user is checked against the supported emergency categories. The system accepts:

- Accident
- Medical
- Fire
- Crime

Invalid emergency types are rejected.

4.2.2 Geographic Data Validation

The latitude and longitude received from the user are converted into numerical values and checked against valid geographical ranges.

Latitude:

$$-90 \leq Latitude \leq 90$$

Longitude:

$$-180 \leq Longitude \leq 180$$

This prevents invalid geographical values from being passed to the routing algorithm.

4.2.3 Road Network Preparation

The road network is loaded into the application using OSMnx and represented as a graph using NetworkX.

The graph contains:

- Nodes representing road locations
- Edges representing road connections
- Edge lengths representing road distance

The nearest graph nodes corresponding to the resource location and emergency location are identified before applying A* search.

4.2.4 Resource Data Preparation

The resource allocation module identifies the resource type required for the selected emergency. It then checks the available resource information and selects an appropriate resource.

4.2.1 ALGORITHMS / METHODOLOGY

The proposed system uses two major classical AI approaches:

1. **Knowledge-Based Reasoning**
2. *A Search Algorithm**

These approaches work together to perform emergency resource allocation and route planning.

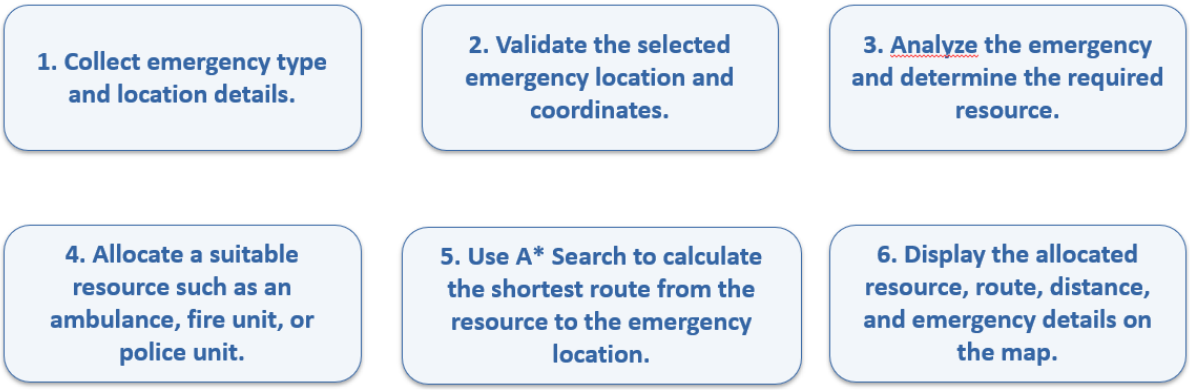


Figure 4.2.1: Methodology

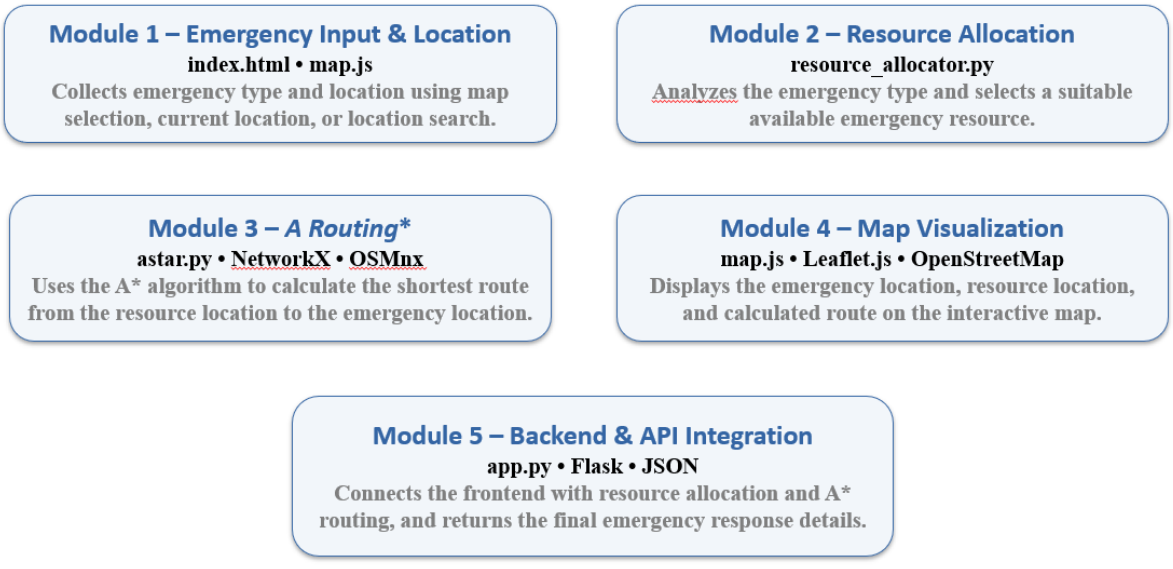


Figure 4.2.1: Methodology Modulus

4.2.1.1 KNOWLEDGE-BASED REASONING

The knowledge-based reasoning module uses predefined IF-THEN rules.

The rules represent the relationship between an emergency type and the required emergency resource.

Example:

IF Emergency = Accident THEN Resource = Ambulance
IF Emergency = Medical THEN Resource = Ambulance
IF Emergency = Fire THEN Resource = Fire Truck
IF Emergency = Crime THEN Resource = Police Vehicle

Figure: 4.2.1.1 Example

The reasoning process can be represented as:



Figure 4.2.1.1: Knownlegde-based reasoning

This approach provides a simple and explainable form of artificial intelligence because the decision can be traced directly to the predefined rule.

4.2.1.2 RESOURCE ALLOCATION PROCESS

The resource allocation process follows these steps:

- The user logs into the system.
- The user selects the emergency type.
- The user provides or selects the emergency location.
- The system validates the input.
- The knowledge base identifies the required resource.
- The system checks resource availability.
- An available resource is selected.
- The selected resource location is obtained.
- The resource and emergency locations are passed to the routing module.
- The allocation result is displayed to the user.
- The allocation activity is recorded.

4.2.1.3 A* SEARCH ALGORITHM

A* is a graph-search algorithm used to find an efficient path between a starting node and a destination node.

The evaluation function used by A* is:

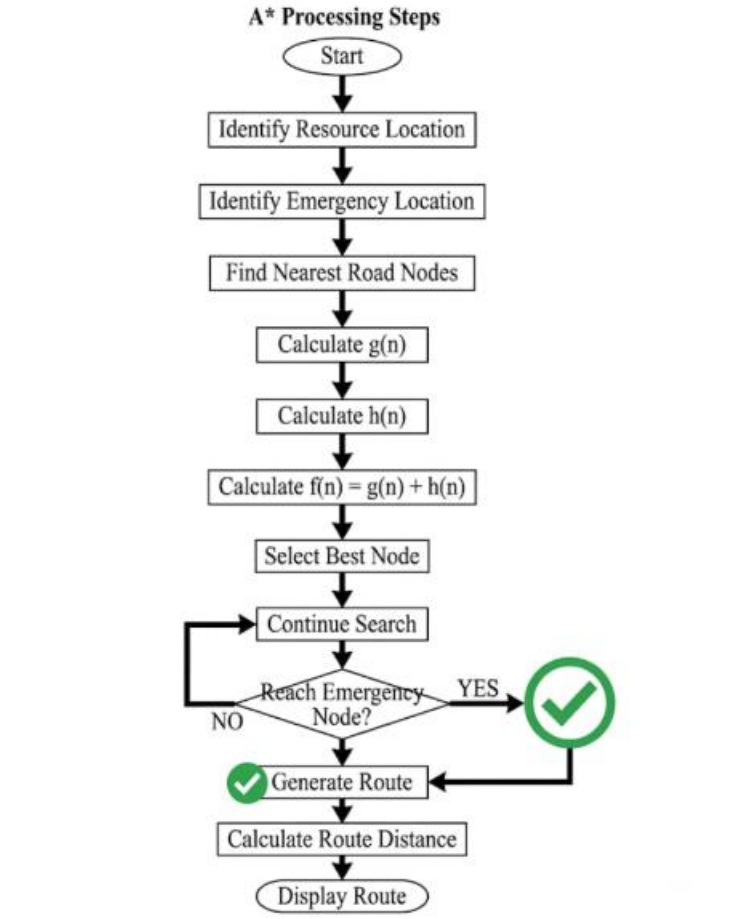
$$f(n) = g(n) + h(n)$$

where:

- $f(n)$is the estimated total cost of the path through node n
- $g(n)$is the actual cost from the starting node to n
- $h(n)$is the heuristic estimate from $n$to the goal

In the proposed system, road distance is used as the path cost.

aphical distance between nodes is used as the heuristic estimate.



A conceptual A* processing steps flow diagram for resource allocation.

Figure 4.2.1.3:A*Search Alogrthim

4.2.1.4 ROUTE DISTANCE CALCULATION

After the A* algorithm determines the route, the system calculates the total route distance using the lengths of the road-network edges forming the selected path.

The total distance is calculated as:

$$D = \sum_{i=1}^{n-1} d_i$$

where:

- D= total route distance
- d_i= length of each road segment
- n= number of nodes in the selected route

The final distance is converted from meters to kilometers for display on the dashboard.

4.2.1.5 SECURITY METHODOLOGY

Security is integrated into the system before the emergency allocation process.

The security workflow is:

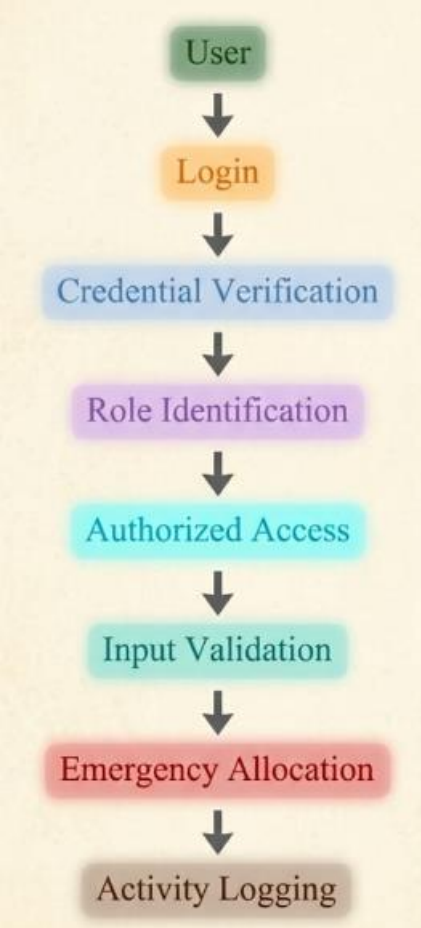


Figure 4.2.1.5: Security Methodology

The system uses authentication to verify users, role-based access control to restrict administrative functions, input validation to reject invalid data, and allocation logging to maintain accountability.

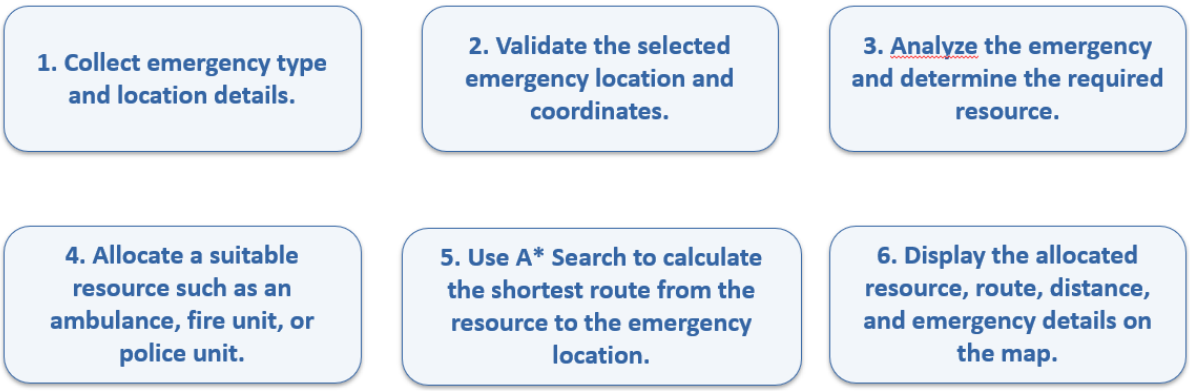


Figure 4.2.1: Methodology

4.2.2 RESULTS AND DISCUSSION

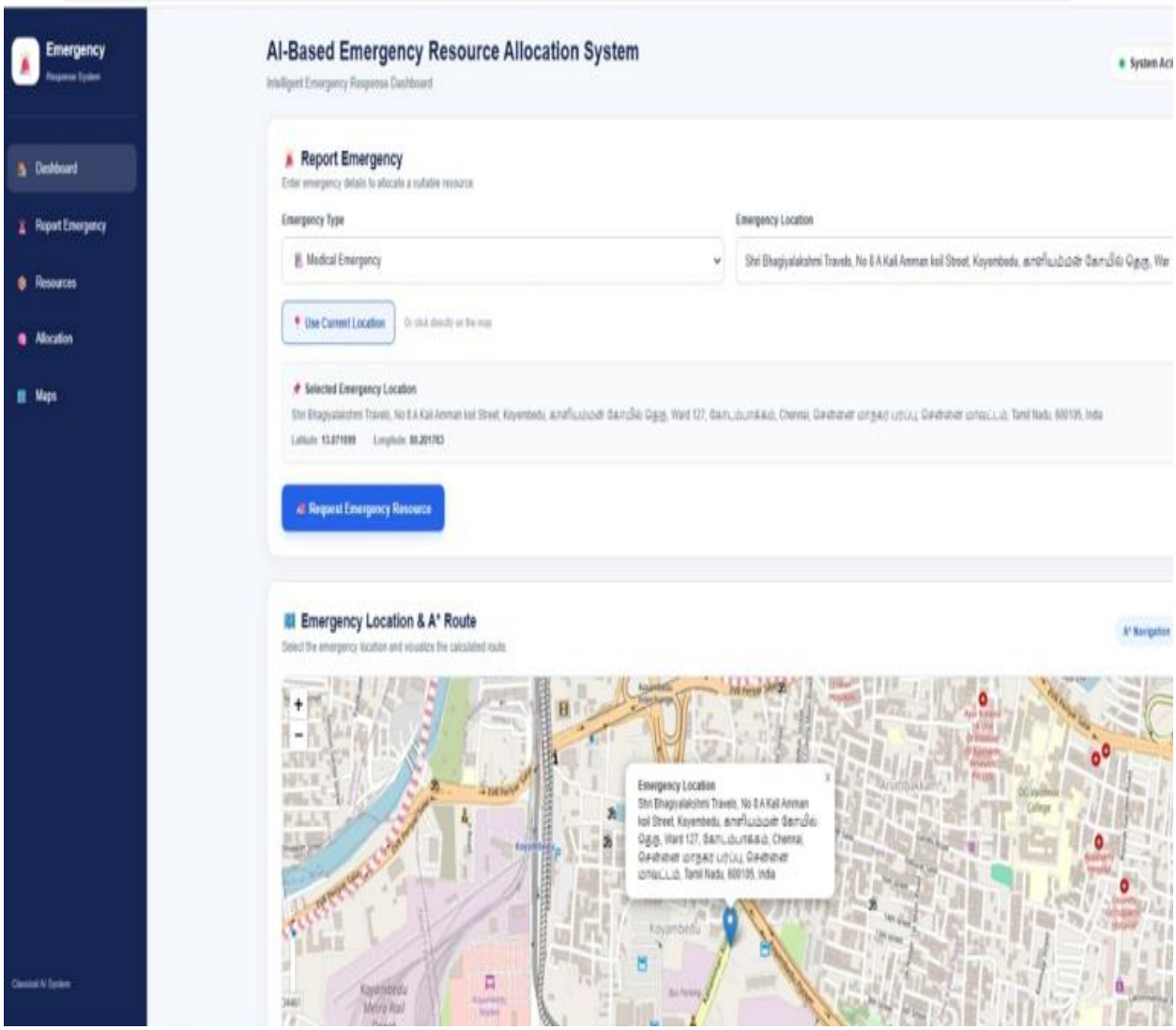


Figure.1.1Emergency Resource Allocation Dashboard

The system processes the request using knowledge-based rules and identifies the corresponding emergency resource. Resource availability is then considered before the allocation is returned.

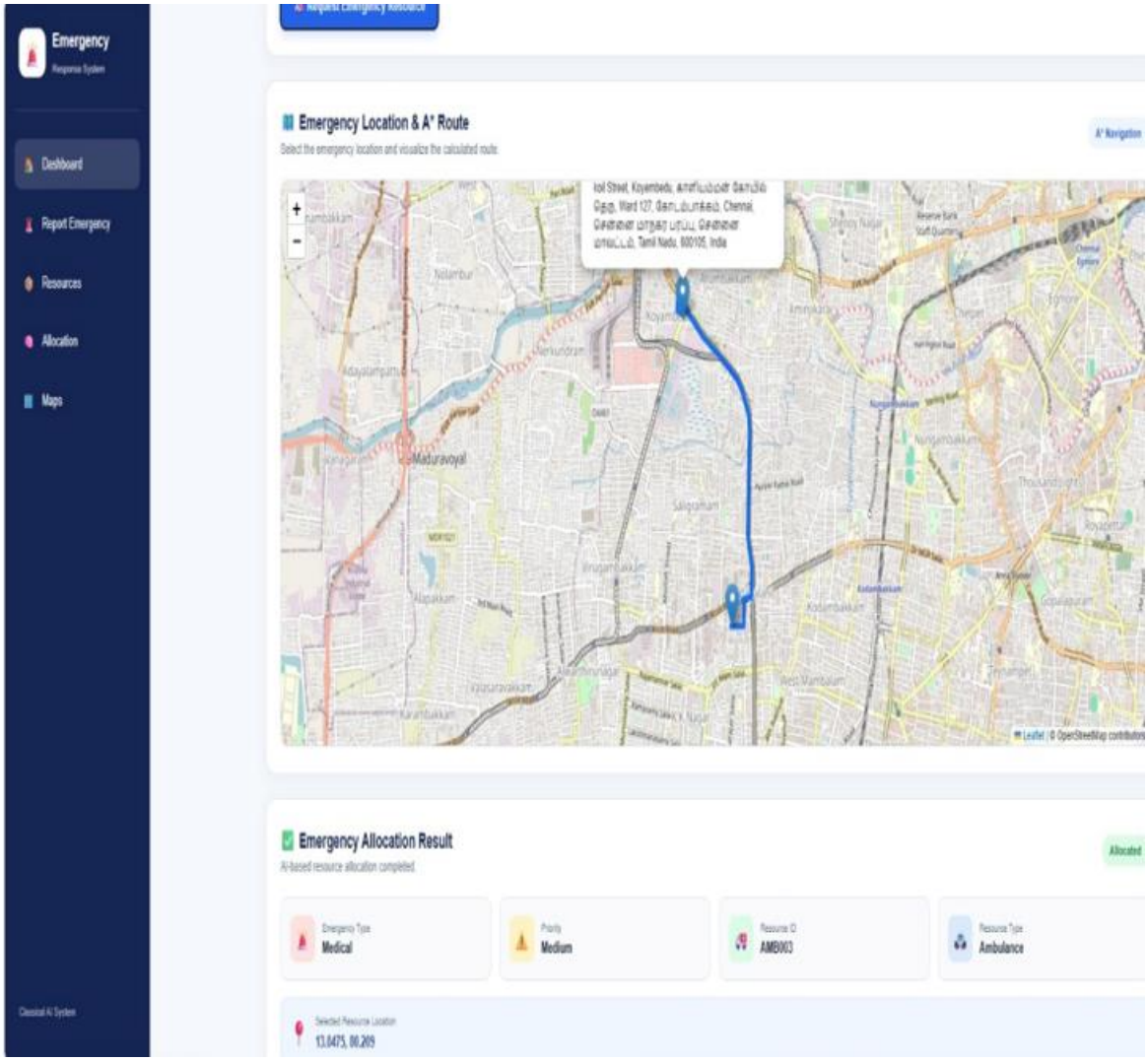


Figure 1.2 Map-Based Emergency Location and Route Visualization

The implemented system provides a web-based interface for emergency resource allocation. After successful authentication, an authorized user can enter an emergency type and location.The system processes the request using knowledge-based rules and identifies the corresponding emergency resource. Resource availability is then considered before the allocation is returned.

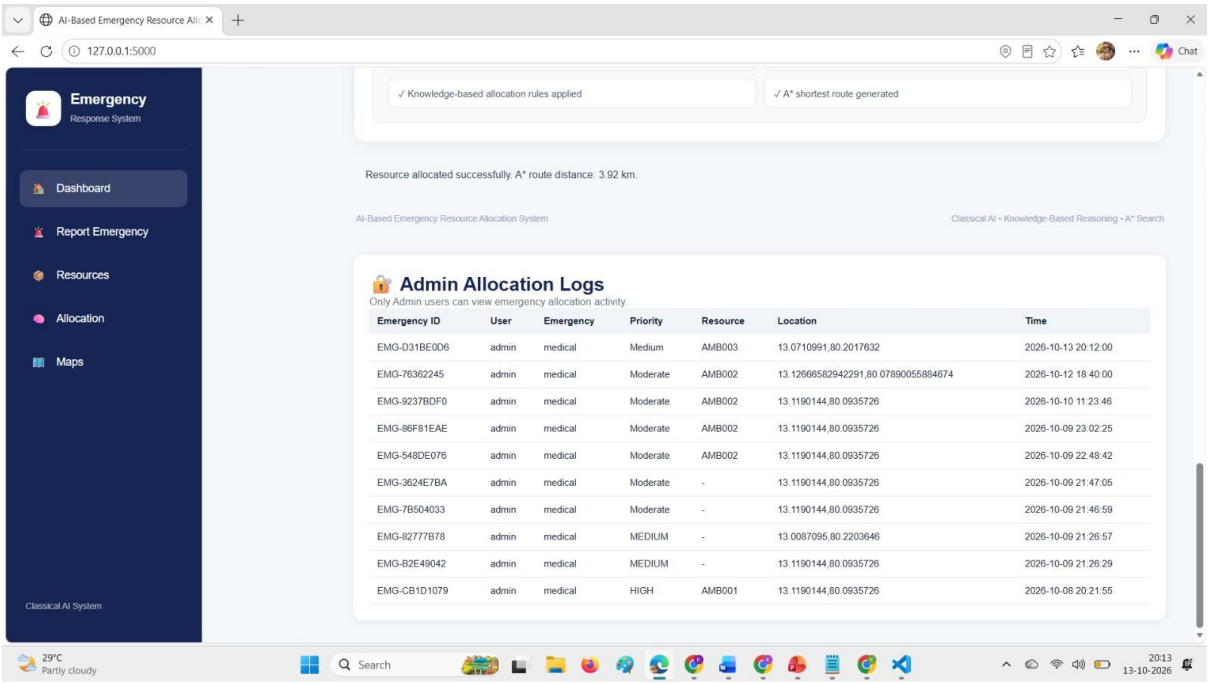


Figure 1.3 Admin Allocation Logs and Resource Monitoring

Once the resource is selected, the A* search module calculates a route between the resource location and the emergency location using the available road network.The resulting information is presented through the dashboard.

4.2.2.1 Authentication Result

The system provides a login page for authorized users.

Two roles are supported:

User Role	Access
Admin	Dashboard and allocation logs
Operator	Emergency allocation and route results

Invalid credentials are rejected, preventing unauthorized access to the protected dashboard.

4.2.2.2 Resource Allocation Result

After an emergency request is submitted, the dashboard displays information about the selected resource.

The result includes:

- Emergency type
- Priority
- Resource ID
- Resource type

- Resource location
- Allocation status

This allows the operator to understand which emergency resource has been selected.

4.2.2.3 A* Route Result

The A* module produces a route between the selected resource and the emergency location.

The dashboard displays:

- Route distance
- Number of route nodes
- Route on the interactive map

For example, during testing, the system successfully generated a route containing **225 nodes** with a route distance of approximately **16.86 km** for a tested Chennai emergency scenario.

The exact route distance depends on the selected resource and emergency coordinates.

4.2.2.4 Map Visualization Result

The calculated route is displayed on the interactive map.

The map provides a visual representation of:

- Emergency location
- Selected resource location
- A* calculated route

This improves the readability of the routing result and helps the operator understand the relationship between the resource and emergency locations.

4.2.2.5 Security Result

The implemented security mechanisms provide the following results:

- Unauthorized users are redirected to the login page.
- Valid users can access the dashboard.
- Admin users can access allocation logs.
- Operator users cannot access Admin-only logs.
- Invalid emergency types are rejected.
- Invalid latitude and longitude values are rejected.
- Allocation activities are recorded.

These controls provide basic authentication, authorization, validation, and accountability for the prototype.

4.2.2.6 Overall System Result

The complete system follows the sequence:

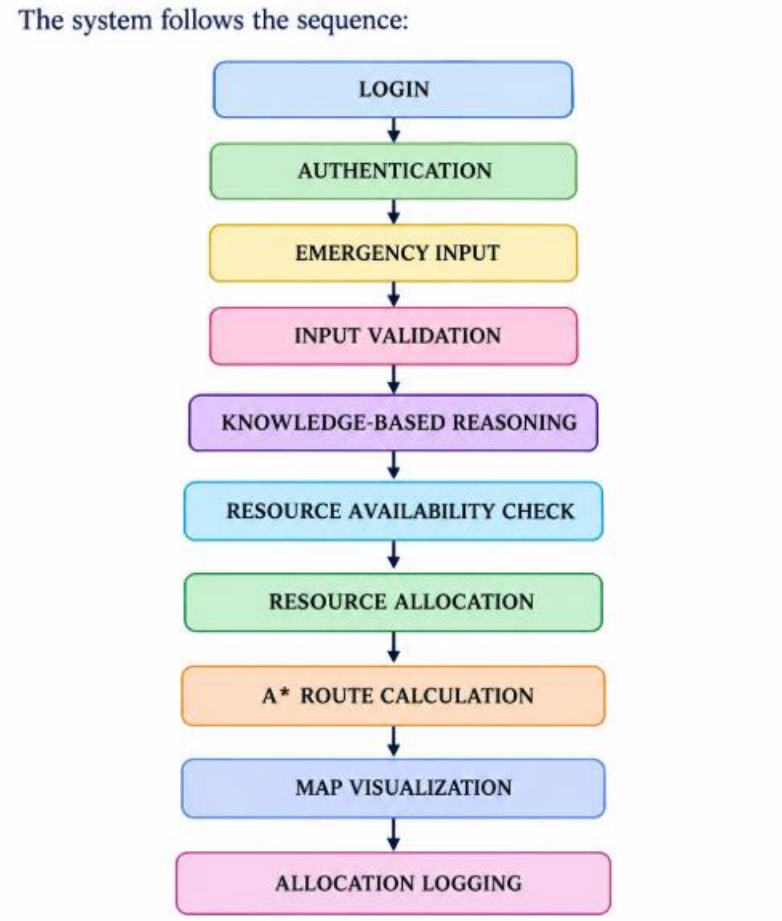


Figure 4.2.2.6 System Result

The results demonstrate that the proposed classical AI approach can combine rule-based emergency resource selection with A* route planning in a single web-based decision-support system.

TEST CASES

You can add a table like this:

Test Case	Input	Expected Output	Status
TC01	Accident + Location	Ambulance allocated	Pass
TC02	Medical + Location	Ambulance allocated	Pass
TC03	Fire + Location	Fire resource allocated	Pass
TC04	Crime + Location	Police resource allocated	Pass
TC05	Invalid emergency type	Error message	Pass
TC06	No location	Location required message	Pass
TC07	Valid location + resource	A* route displayed	Pass
TC08	Location search	Search results displayed	Pass

Figure 4.2.2.7 Result Test Case

5. CONCLUSION

The AI Emergency Resource Allocation System demonstrates the application of classical Artificial Intelligence techniques in emergency response decision support.The system combines a knowledge-based resource allocation approach with the A* Search Algorithm to select an appropriate emergency resource and determine an efficient route to the emergency location.The system provides an interactive web interface using HTML, CSS, JavaScript, Leaflet.js, and OpenStreetMap. The Flask backend manages emergency requests and connects the resource allocation and routing modules. NetworkX and OSMnx are used for graph-based route calculation and road network processing.The system successfully demonstrates how classical AI can be applied to emergency resource allocation and route planning. The proposed approach provides a foundation for future development of more advanced emergency response systems.

Future Enhancements

- Real-time traffic information.
- Live GPS tracking of emergency vehicles.
- Multiple resource comparison.
- Dynamic resource availability.
- Real-time emergency notifications.
- Machine learning-based emergency prediction.
- Cloud deployment.
- Mobile application support.
- Authentication and authorization.
- Real-time traffic-aware A* routing**.**

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[2]A. Mukhopadhyay, G. Pettet, S. M. Vazirizade, D. Lu, A. Jaimes, S. E. Said, H. Baroud, Y.Vorobeychik, M. J. Kochenderfer, and A. Dubey, "A Review of Incident Prediction, Resource Allocation, and Dispatch Models for Emergency Management," in Accident Analysis & Prevention, vol. 165, Art. no. 106501.

[3]G. Pettet, A. Mukhopadhyay, M. J. Kochenderfer, and A. Dubey, "Hierarchical Planning for Resource Allocation in Emergency Response Systems," in Proceedings of the 12th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS 2021), Nashville, TN, USA, May

APPENDIX

APPENDIX A – TECHNOLOGY STACK

Category	Technology
Programming Language	Python
AI Technique	Classical AI, Rule-Based Reasoning
Search Algorithm	A* Search Algorithm
Heuristic Function	Estimated Distance to Destination
Emergency Data Collection	HTML Forms
Input Validation	Python Validation Rules
Priority Determination	Rule-Based Priority Assignment
Resource Availability	JSON-Based Resource Management
Resource Allocation	Rule-Based Resource Selection
Backend	Flask
Frontend	HTML, CSS, JavaScript
Data Storage	JSON Files
Map Visualization	Route and Location Display
Development Environment	Visual Studio Code
Testing	Manual Testing, Pytest (if implemented)

APPENDIX B – PROJECT STRUCTURE

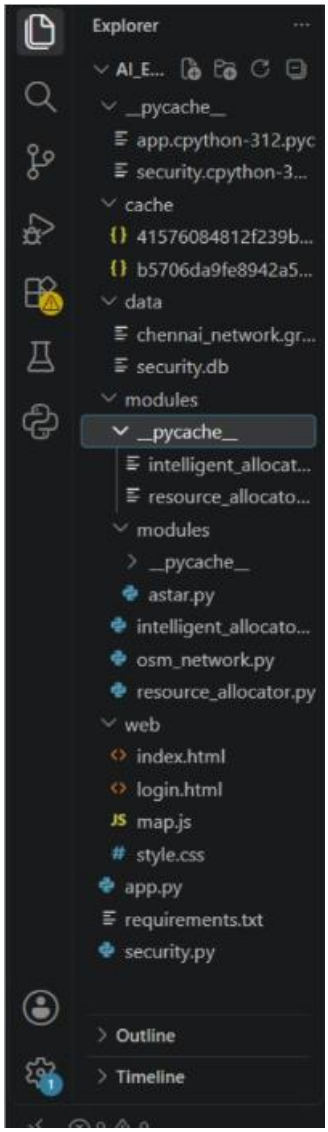


Figure Structure

APPENDIX C – SCREENSHOTS

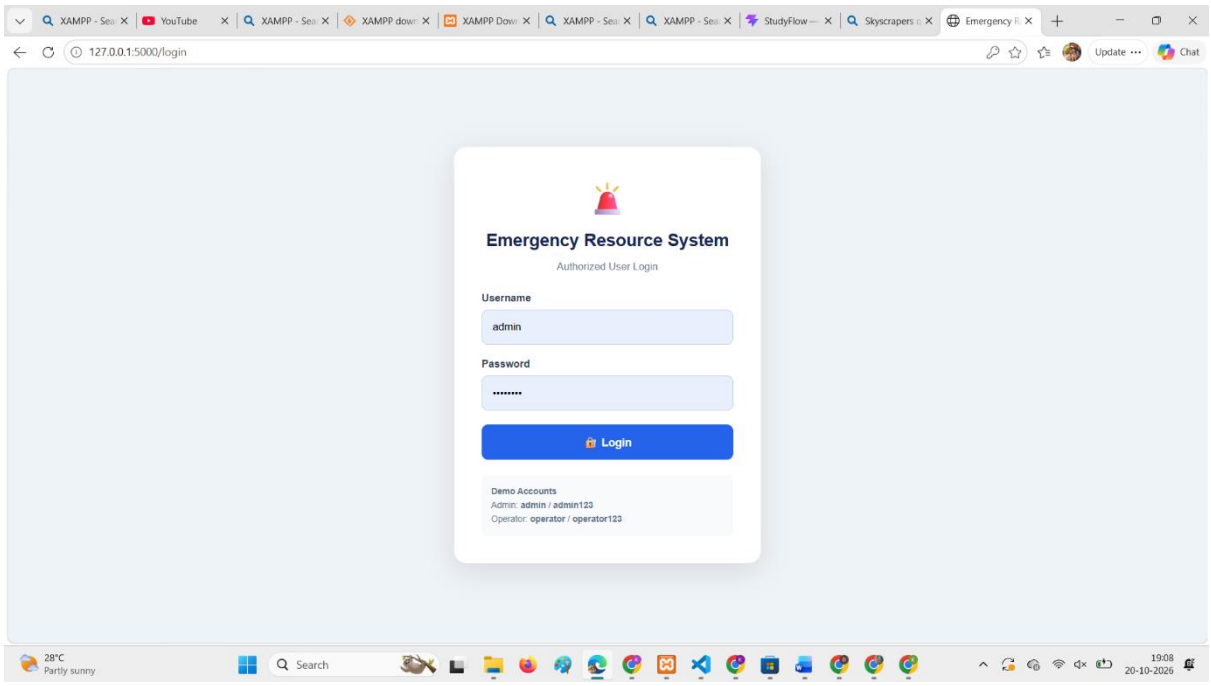


Figure Admin Dashboard

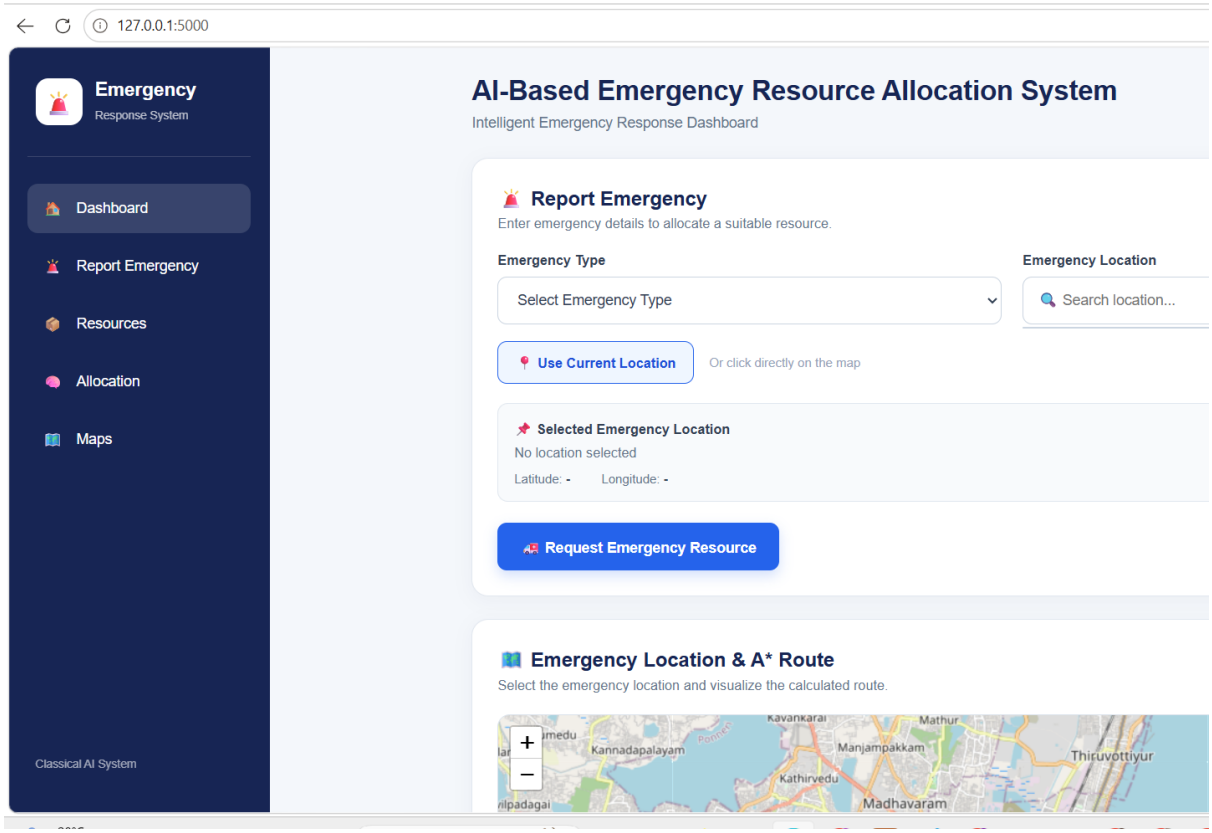


Figure Allocation Dashboard

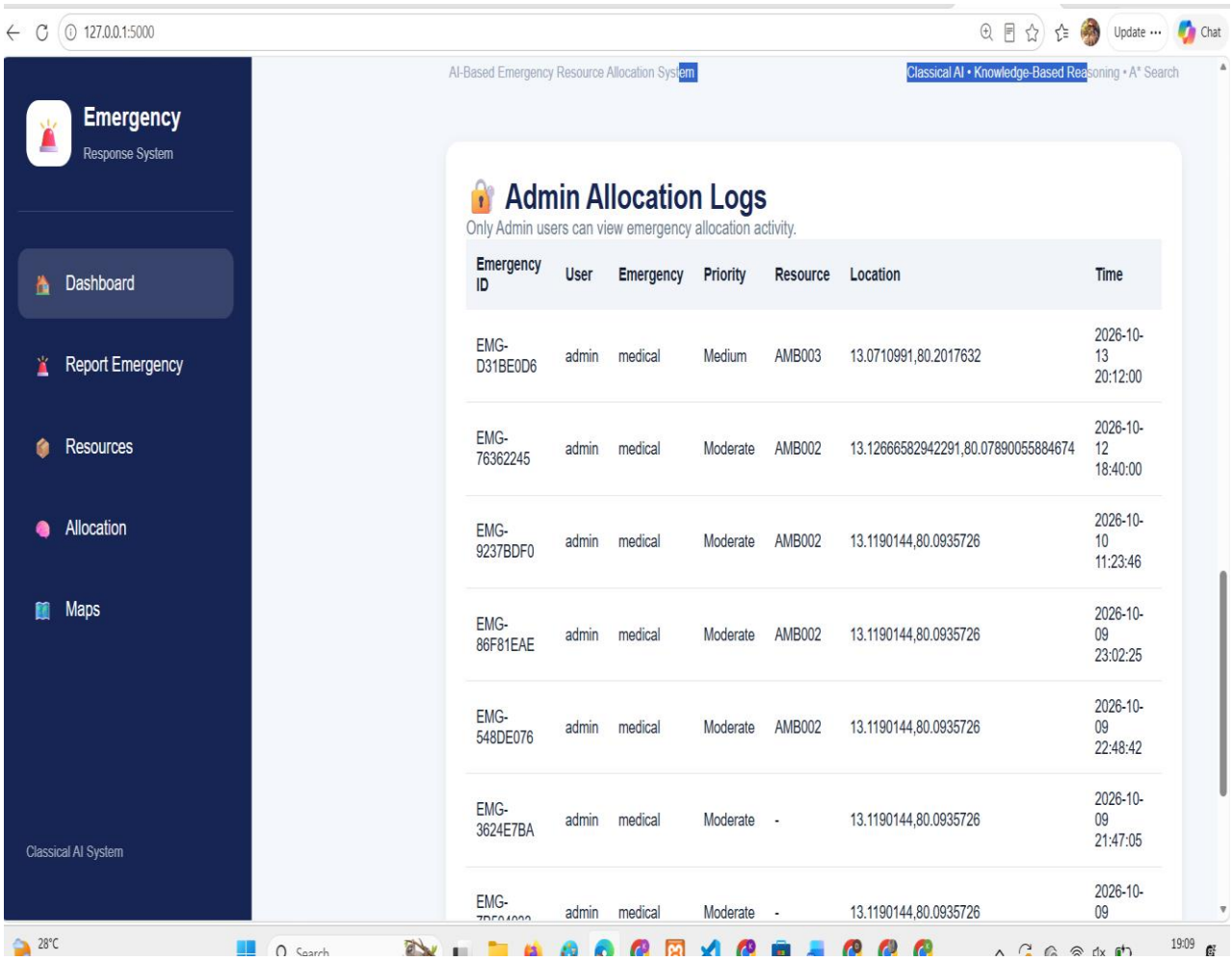


Figure Admin Allocation Logs and Resource Monitoring

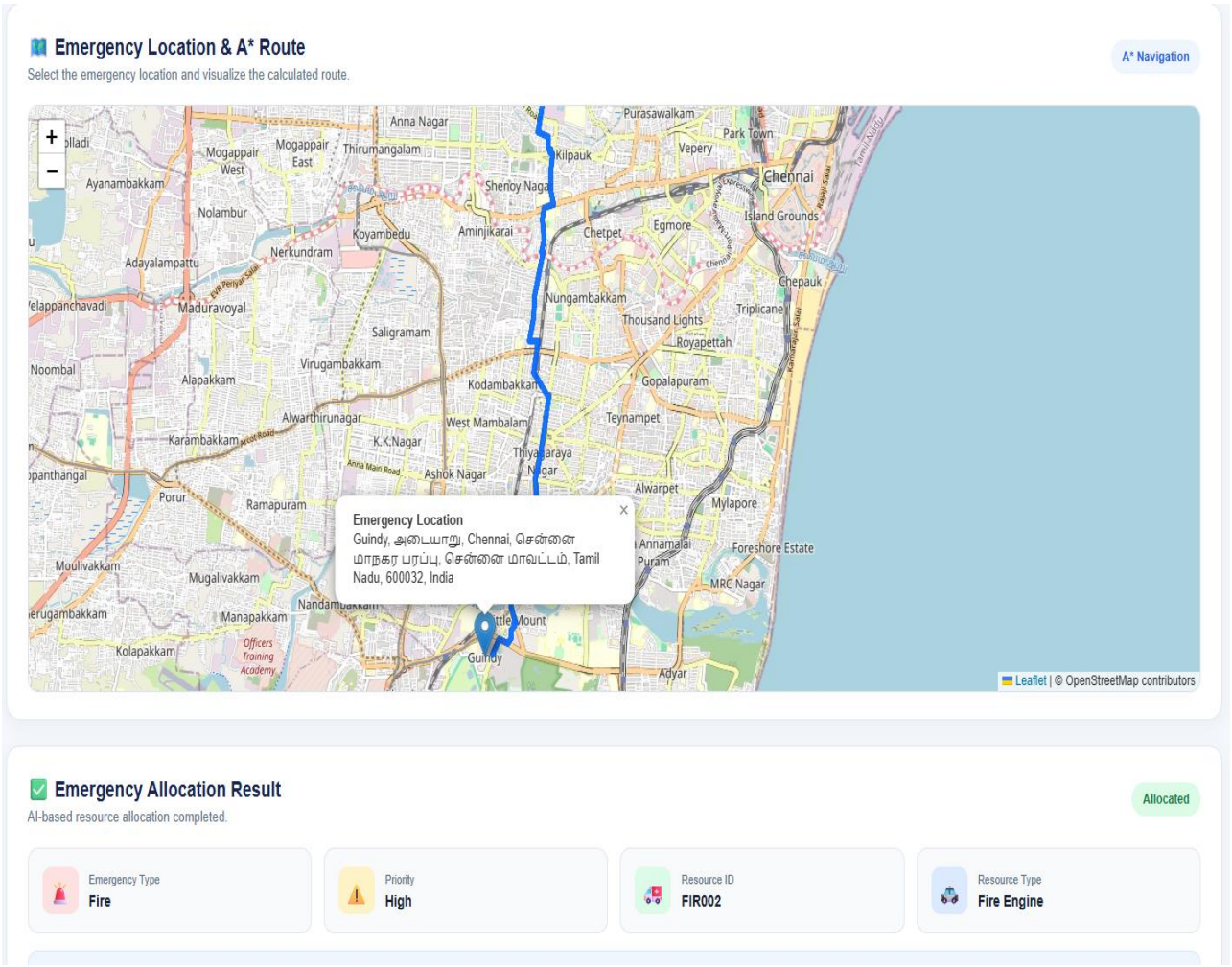


Figure Map-Based Emergency Location and Route Visualization